Two-loop $gg \to AA$ amplitudes near threshold and the double parton scattering singularity

Arunima Bhattacharya *
Theory Division, IFIC, University of Valencia–CSIC, E-46980 Paterna, Valencia, Spain

We study the two-loop Type II amplitudes for pseudoscalar pair production, $gg \to AA$, near threshold in the heavy top quark limit, where each pseudoscalar is emitted from its own effective gluon vertex. Their analytic forms contain large cancellations near threshold. We expand them in the pseudoscalar velocity β_A through β_A^6 at fixed angle. After the endpoint constants are reduced, all inverse powers of β_A and the odd powers through β_A^5 cancel, and every coefficient is even in $\cos\theta$. At leading power the angle enters only through the logarithm of the transverse momentum p_T. We identify the one-loop logarithm with the double parton scattering singularity: with the overall constant fixed at threshold, the factorised formula of Gaunt and Stirling reproduces the velocity dependence of its coefficient in the forward limit to 3×10^{-15}, and a one-loop $gg \to HH$ calculation confirms the CP-even weight with the same constant. At two loops, the equal-helicity amplitude grows as $\ln^3 p_T$, and its logarithms follow the one-loop helicity weight in the forward limit at all velocities tested; the opposite-helicity amplitude, with no one-loop logarithm, grows as $\ln^2 p_T$, and fits relate this term to the one-loop amplitude. In the finite virtual corrections, the one-loop logarithm does not interfere with the tree amplitude, and at two loops the highest logarithm, relative to the squared tree amplitude, is $-\frac{7}{64}\alpha_s^2 N_c^2 \ln^2(2p_T/m_A)$. The released remainders are normalised differently at different orders (a factor 64 at two loops, one-loop files at $\mu_R = 1$ GeV); I convert them to a common normalisation, checked with the renormalisation group equations. For $0 < \beta_A \leq 0.2$, the series reproduce the complete remainders to 6.77×10^{-5} at all tested points; they are released with an evaluator.

I. INTRODUCTION

Precision tests of the Standard Model (SM) and searches for new particles at the Large Hadron Collider (LHC) require theoretical predictions for production rates and differential cross sections. The accuracy of these predictions is important when comparing different models or determining whether a discrepancy with a measurement indicates a new interaction. QCD radiative corrections in gluon fusion processes can be sizeable, and calculations beyond leading order (LO) are needed to assess their effect on the predicted rates and distributions [1]. Loop amplitudes enter the virtual contributions to these corrections. Their calculation requires the evaluation of loop integrals and control over their behaviour in the physical region. In particular, amplitudes can develop singularities at special kinematic configurations even when the corresponding tree amplitudes are regular. The location and strength of these singularities matter for numerical evaluation and for the behaviour of differential cross sections [2, 3]. An expansion about such a configuration allows the singular terms to be studied explicitly, together with the regular terms needed for a quantitative prediction.

A relevant example is neutral Higgs boson pair production in two-Higgs-doublet models (2HDM). At the LHC, the dominant production mechanism is gluon fusion through top quark loops, so that loop amplitudes enter already at LO [1, 4]. For pseudoscalar pairs, $gg \to AA$, the next-to-leading order (NLO) QCD corrections were first computed in the limit of a heavy top quark [5], later matched to parton showers [6], and subsequently obtained with the full dependence on the top quark mass [1]. In the benchmark considered in Ref. [1], the full NLO cross section for AA lies about 5% below the Born-improved heavy top quark approximation at LHC energies, compared with about 12% for hH. For this benchmark, the comparison supports using the heavy top quark limit (HTL) as a starting point for higher orders, with the exact mass dependence retained at LO. The two-loop Type II finite remainders in this limit were obtained in Ref. [7]. For SM Higgs pairs, the corrections in the same limit are known to N^{3}LO [8–10]. The same effective gluon interactions also contribute to the production of axion-like particle pairs, and non-resonant $gg \to aa$ production has been used to constrain their interactions at the LHC [11].

In the limit of a heavy top quark, a pseudoscalar field $\Phi^A(x)$ couples to gluons and light quarks through the effective Lagrangian [12]

$$\mathcal{L}^A_{\text{eff}} = \Phi^A(x)\left[-\frac{1}{8}C_G\, O_G(x) - \frac{1}{2}C_J\, O_J(x)\right], \quad (1)$$

with the gluonic and light quark operators

$$O_G(x) = G^{a\mu\nu}\tilde{G}^a_{\mu\nu} = \epsilon_{\mu\nu\rho\sigma}G^{a\mu\nu}G^{a\rho\sigma}, \quad (2)$$

$$G^{a\mu\nu} = \partial^\mu G^{a\nu} - \partial^\nu G^{a\mu} + g_s f^{abc} G^\mu_b G^\nu_c, \quad (3)$$

where f^{abc} are the structure constants of SU(N_c), with $N_c = 3$ in QCD, and $\epsilon_{\mu\nu\rho\sigma}$ is the Levi-Civita tensor. The light quark operator is the divergence of the flavour

* arunimab@ific.uv.es

singlet axial vector current,

$$O_J(x) = \partial_\mu \left(\bar{\psi}\gamma^\mu\gamma^5\psi\right) . \tag{4}$$

By the Adler–Bardeen theorem [13, 14], C_G receives no QCD corrections beyond one loop, whereas C_J starts at second order in the strong coupling $a_s \equiv g_s^2/16\pi^2 = \alpha_s/4\pi$. The Wilson coefficients are

$$C_G\left(a_s\right) = -a_s 2^{\frac{5}{4}} G_F^{\frac{1}{2}} \cot\beta\,, \tag{5}$$

$$C_J\left(a_s\right) = -\left[a_s C_F\left(\frac{3}{2} - 3\ln\frac{\mu_R^2}{m_t^2}\right) + a_s^2 C_J^{(2)} + ...\right] C_G\,, \tag{6}$$

where G_F is the Fermi constant, m_t the top quark mass and $\tan\beta$ the ratio of the vacuum expectation values of the two Higgs doublets, in a model without spontaneous CP violation; this angle is unrelated to the velocity

$$\beta_A = \sqrt{1 - 4m_A^2/\hat{s}} \tag{7}$$

of the pseudoscalars in the partonic centre of mass frame, at fixed scattering angle. C_F is the quadratic Casimir of the fundamental representation and μ_R the scale at which a_s is renormalised. The conventions are those of Ref. [7]. Operator mixing, ultraviolet renormalisation and the treatment of γ_5 [15] follow the original calculation in Ref. [7]; the same effective theory underlies inclusive pseudoscalar production at NNLO [16].

We consider the dominant gluon fusion subprocess

$$g(p_1) + g(p_2) \to A(p_3) + A(p_4), \tag{8}$$

with $p_1^2 = p_2^2 = 0$ for the incoming gluons and $p_3^2 = p_4^2 = m_A^2$ for the outgoing pseudoscalars. The amplitude for $gg \to AA$ separates into two classes [7]. In Type I, both pseudoscalars originate from a single effective vertex. In Type II, each pseudoscalar is emitted from its own vertex and the two vertices are connected by gluon exchange in the t or u channel (Fig. 1a). In the full theory, the Type II tree diagrams correspond to two top quark triangles joined by a gluon, the one-particle reducible contributions of Refs. [1, 5]; for Higgs pairs the corresponding class was computed at two loops [9]. The two-loop Type II finite remainders for $gg \to AA$ were obtained in closed form in Ref. [7]. They contain polylogarithms of weight up to four, among them $\mathrm{Li}_{2,2}$, and inverse powers of the Gram determinant $tu - m_A^4$, which vanishes at the pair threshold. The expressions occupy about 204 MB, and large cancellations between their terms make them hard to evaluate close to threshold.

There are two reasons to study this challenging threshold region. The first is phenomenological: in the NLO predictions of Ref. [1], for $m_A = 134.7$ GeV the invariant mass distribution of the pair is larger at $m_{AA} = 300$ GeV, about 30 GeV above threshold, than at 400 GeV, and threshold resummation of this distribution requires at each m_{AA} a hard function obtained from the infrared (IR) finite virtual amplitude [17–21]. The point $m_{AA} = 300$ GeV corresponds to $\beta_A \approx 0.44$; the threshold series of this paper is accurate for $\beta_A \leq 0.2$, that is $m_{AA} \lesssim 1.02 \times 2m_A$, while the forward-limit tests of Sec. IV D concern selected logarithmic coefficients of the complete amplitude at larger velocities.

The second concerns the analytic structure of loop amplitudes. When the transverse momentum p_T of the final state particles vanishes, a one-loop amplitude with massless internal lines can have a Landau singularity [22] in which each incoming parton splits into two collinear partons that then take part in two separate $2 \to 1$ scatterings. Gaunt and Stirling showed that this double parton scattering (DPS) singularity is at most logarithmic in p_T for SM amplitudes and that, for the processes they considered, it occurs only when the incoming state has $J_z = 0$. They also derived a factorised formula for its coefficient [2]. Dixon and Esterlis studied the corresponding singularity of six-gluon amplitudes in planar $\mathcal{N} = 4$ super Yang–Mills theory to high loop order [3]. However, it has not been examined beyond one loop in QCD at the level of amplitudes for a $2 \to 2$ process with a colour singlet final state; NLO corrections to the underlying splitting mechanism are known [23] for double parton distributions [24]. The availability of a complete multiloop amplitude does not by itself make the behaviour of a particular Landau region explicit: cancellations between kinematic prefactors and logarithms can obscure the singular limit. The threshold expansion below isolates this structure in an explicit two-loop QCD amplitude. The Type II amplitude for $gg \to AA$ contains diagrams with this topology: each incoming gluon can split into two gluons, which fuse pairwise into the two pseudoscalars at the effective vertices (Fig. 1b). At threshold, $p_T \to 0$ at every scattering angle, so a threshold expansion probes this configuration directly.

In hadronic collisions, a configuration in which a parton from each proton splits into two can enter both the single and double parton scattering calculations. The overlap must therefore be subtracted to avoid counting the same contribution twice [23, 25]. DPS also receives a contribution in which only one of the two parton pairs arises from a perturbative splitting, while the other proton supplies two partons without such a splitting. This contribution has been studied for Higgs boson pairs [26]. DPS has also been observed in same-sign W pair production at the LHC [27, 28]. The overlap between single and double parton scattering has been re-examined using two-loop splitting kernels [29]. The amplitude-level singularity studied here is not itself a double parton distribution or a hadronic DPS cross section. It is the corresponding singular region of the single parton scattering loop amplitude, and therefore provides complementary information on the perturbative splitting configuration that also enters the overlap between single and double parton scattering.

In this paper, we expand the two-loop Type II finite remainders of Ref. [7] in the velocity β_A through β_A^6. The

main results are the following:

1. After the constants that arise in the expansion are expressed in a basis of eight transcendental constants, all inverse powers of β_A and the odd powers through β_A^5 cancel, and every coefficient is even in $\cos\theta$, as required by Bose symmetry. At leading power, the angle enters only through the logarithm of the transverse momentum p_T of the pseudoscalars.

2. This logarithm is the DPS singularity. At one loop its coefficient follows the factorised formula of Ref. [2] for gluon splitting and the opposite signs of the amplitudes for $g^+g^+ \to A$ and $g^-g^- \to A$: with the overall constant fixed at threshold, it reproduces the velocity dependence of the exact one-loop coefficient in the forward limit. A one-loop calculation of $gg \to HH$ confirms the corresponding weight for a CP-even coupling.

3. At two loops the $J_z = 0$ amplitude, with equal gluon helicities, grows as $\ln^3 p_T$, and the ratio of its leading two- and one-loop logarithms agrees with the corresponding result in planar $\mathcal{N} = 4$ super Yang–Mills theory [3]; in the forward limit, its two-loop logarithms follow the one-loop helicity weight at all velocities tested. The $|J_z| = 2$ amplitude has no DPS logarithm at one loop but develops a $\ln^2 p_T$ term at two loops. Forward-limit fits provide numerical evidence that the real coefficient of this double logarithm is $-4N_c$ times the one-loop forward amplitude over the velocities tested; its dynamical origin remains open.

4. In the finite virtual corrections, the one-loop DPS logarithm does not contribute to the interference of the one-loop and tree amplitudes, because it is out of phase with the tree amplitude. At two loops the cubic logarithm is absent, and the leading singular term is

$$V_2 \supset -\frac{7}{4}\pi^2 N_c^2 \ln^2\left(\frac{2p_T}{m_A}\right).$$

   Here V_2 is the two-loop finite virtual coefficient normalised to the squared Type II tree amplitude, as defined in Sec. V. It results from the combination of the interference of the two-loop amplitude with the tree amplitude and the square of the one-loop amplitude.

The remainders released with Ref. [7] are normalised differently at different orders, with a factor 64 in the two-loop files and the one-loop files given at $\mu_R = 1$ GeV; we convert them to a common normalisation (App. B). The series reproduce the unexpanded remainders with relative errors of at most 6.77×10^{-5} for $\beta_A \leq 0.2$; they are released together with an evaluator [30].

The paper is organised as follows. Section II sets out the conventions, the normalisation of the remainders and the counting of orders. Section III summarises the procedure and describes the expansion and its accuracy. Section IV discusses the leading power terms and their relation to the DPS singularity, and Sec. V their consequences for the cross section. The conclusions are in Sec. VI.

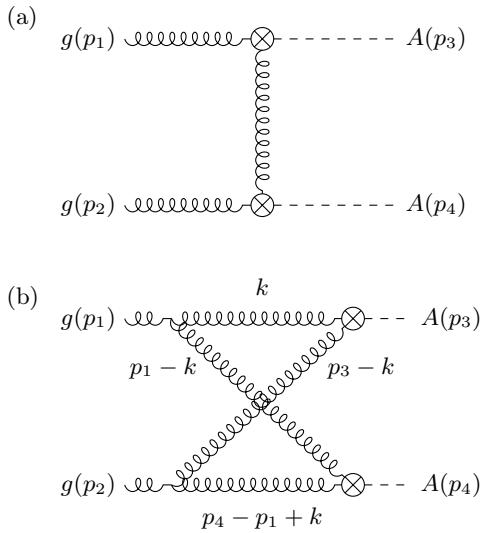


FIG. 1. (a) Type II tree diagram in the t channel; the u-channel diagram is obtained by exchanging the pseudoscalars. The crossed circles are the effective vertices of O_G. (b) One-loop Type II diagram with loop momentum k. Its double parton scattering singularity lies at $k = xp_1$ and $p_3 - k = (1-x)p_2$, where the four internal gluons are on shell and collinear with the incoming gluons: each incoming gluon splits into two, which fuse pairwise at the effective vertices. Gaunt and Stirling call this topology the crossed box [2].

II. AMPLITUDES AND CONVENTIONS

A. Theoretical framework

To study the threshold behaviour of $gg \to AA$ through two loops, we use the amplitudes from Ref. [7] with the effective Lagrangian of Eq. (1). The labels Type I and Type II follow that reference and distinguish the two classes of diagrams. They are unrelated to the Yukawa types of 2HDMs. The Type II tree diagrams correspond, in the full theory, to two top quark triangles connected by a gluon, i.e. the one-particle reducible contribution of Ref. [1]. Here we consider the contribution with two O_G insertions and omit the superscript II from the amplitudes. Diagrams with one O_G and one O_J insertion first contribute at one loop. Their coefficients are proportional to the O_G–O_G tree coefficients, as discussed in Sec. II D.

B. Kinematics

The Mandelstam variables are

$$\hat{s} = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_2 - p_3)^2, \quad (9)$$

with $\hat{s} + t + u = 2m_A^2$. Following the conventions of Ref. [7], we use

$$\hat{s} = m_A^2 \frac{(1+x)^2}{x}, \quad t = -m_A^2 y, \quad u = -m_A^2 z, \quad (10)$$

with $x + x^{-1} = y + z$. With $c = \cos\theta$ in the partonic centre of mass frame,

$$\hat{s} = \frac{4m_A^2}{1-\beta_A^2}, \qquad t, u = m_A^2 - \frac{\hat{s}}{2}(1 \mp \beta_A c), \quad (11)$$

Solving Eq. (11) for the variables of Eq. (10) and taking the branch $0 < x \leq 1$ gives the threshold substitution

$$x = \frac{1-\beta_A}{1+\beta_A}, \qquad y, z = \frac{1+\beta_A^2 \mp 2\beta_A c}{1-\beta_A^2}. \quad (12)$$

All three variables tend to one as $\beta_A \to 0^+$ at fixed c, and the expansion has to follow this correlated limit. The Gram determinant and the transverse momentum are related by $tu - m_A^4 = \hat{s}\, p_T^2$, with

$$tu - m_A^4 = \frac{4m_A^4 \beta_A^2 (1-c^2)}{(1-\beta_A^2)^2}, \qquad p_T^2 = \frac{m_A^2 \beta_A^2 (1-c^2)}{1-\beta_A^2}. \quad (13)$$

Both vanish at threshold at every angle, and also at fixed β_A when $c \to \pm 1$.

C. Tensor decomposition and helicity amplitudes

Every Type II diagram carries two effective vertices, so the amplitude is proportional to C_G^2 at every order. Extracting this factor, we decompose

$$\mathcal{M}_{ab}^{\mu\nu} = C_G^2\, \delta_{ab} \left(T_1^{\mu\nu} \mathcal{M}_1 + T_2^{\mu\nu} \mathcal{M}_2 \right), \quad (14)$$

with

$$T_1^{\mu\nu} = g^{\mu\nu} - \frac{p_1^\nu p_2^\mu}{p_1 \cdot p_2}, \quad (15)$$

$$T_2^{\mu\nu} = g^{\mu\nu} + \frac{1}{(p_1\cdot p_2) p_T^2} \Big[m_A^2 p_2^\mu p_1^\nu - 2(p_1\cdot p_3) p_2^\mu p_3^\nu - 2(p_2\cdot p_3) p_3^\mu p_1^\nu + 2(p_1\cdot p_2) p_3^\mu p_3^\nu \Big], \quad (16)$$

$$p_T^2 = \frac{tu - m_A^4}{\hat{s}}. \quad (17)$$

For physical gluon polarisations, T_1 contributes only to equal helicities ($J_z = 0$) and T_2 only to opposite helicities ($|J_z| = 2$), each with a contraction of unit modulus. Up to helicity-dependent overall phases, which drop out of every squared quantity, the amplitudes for incoming gluon helicities ($\lambda_1 \lambda_2$) are [31]

$$\mathcal{A}_{\pm\pm} = C_G^2\, \mathcal{M}_1, \qquad \mathcal{A}_{\pm\mp} = C_G^2\, \mathcal{M}_2\, e^{\pm 2i\phi}, \quad (18)$$

where ϕ is the azimuthal angle of $\vec{p}_3$ about the beam axis. Equal helicities correspond to $J_z = 0$ along the beam and opposite helicities to $J_z = \pm 2$; the phase of $\mathcal{A}_{\pm\mp}$ is fixed by rotations about the beam axis. Summing over the two configurations of each kind and over colour,

$$\sum_{\text{hel,col}} |\mathcal{A}|^2 = 2(N_c^2 - 1)\, C_G^4 \left(|\mathcal{M}_1|^2 + |\mathcal{M}_2|^2 \right). \quad (19)$$

The labels 1 and 2 refer to the tensors of Eq. (14), not to the Type I and Type II diagram classes.

At tree level the two Type II diagrams give

$$\mathcal{M}_1^{(0)} = \frac{i\hat{s}}{2},$$
$$\mathcal{M}_2^{(0)} = \frac{i m_A^2 (x-z)(x^2+1)(xz-1)}{4xz(x^2 - xz + 1)} = 2i m_A^2 \beta_A^2 (1-c^2) + \mathcal{O}(\beta_A^4), \quad (20)$$

in the variables of Eq. (10). We obtained Eq. (20) from the projections of the original calculation and checked it against a direct evaluation of the two Feynman diagrams with the vertex $iC_G \varepsilon_{\mu\alpha\rho\beta} k_1^\mu k_2^\rho \delta^{ab}$ that follows from Eq. (1) (App. B). The suppression of $\mathcal{M}_2^{(0)}$ by β_A^2 is the centrifugal barrier of the $|J_z| = 2$ state: two spin-zero particles can carry two units of angular momentum along the beam only orbitally, at a cost of β_A^2 near threshold.

D. Finite remainders and normalisation

The coefficients $\mathcal{M}_i$ of Eq. (14), from which C_G^2 has been factored out, are expanded as

$$\mathcal{M}_i = \mathcal{M}_i^{(0)} + a_s \mathcal{M}_i^{(1)} + a_s^2 \mathcal{M}_i^{(2)} + \cdots, \quad (21)$$

with a_s renormalised at μ_R. We use dimensional regularisation with $d = 4 + \epsilon$, following Ref. [7]. The finite remainders, called the finite part in Ref. [7], are defined by the IR subtraction of Refs. [7, 32],

$$\mathcal{M}^{(1)} = 2I^{(1)} \mathcal{M}^{(0)} + \mathcal{M}^{(1),\text{fin}},$$
$$\mathcal{M}^{(2)} = 4I^{(2)} \mathcal{M}^{(0)} + 2I^{(1)} \mathcal{M}^{(1)} + \mathcal{M}^{(2),\text{fin}}. \quad (22)$$

From here on $\mathcal{M}^{(n)}$ denotes the finite remainder $\mathcal{M}^{(n),\text{fin}}$.

At two loops, we decompose each finite remainder into powers of the number of light quark flavours,

$$\mathcal{M}_i^{(2)} = \mathcal{M}_{i,0}^{(2)} + n_f\, \mathcal{M}_{i,1}^{(2)} + n_f^2\, \mathcal{M}_{i,2}^{(2)}. \quad (23)$$

Each $\mathcal{M}_{i,k}^{(2)}$ is finite, but none is the amplitude of a separate process: the term linear in n_f, for example, combines the diagrams with a closed quark loop with the terms of

the renormalisation and of the infrared subtraction that depend on n_f. In the rest of the paper, the complete coefficient means $\mathcal{M}_i^{(2)}$ at $n_f = 5$. Furthermore, the remainders released with Ref. [7] are normalised differently at different orders: the two-loop remainders carry an overall factor 64 relative to the tree and one-loop amplitudes, and the one-loop remainders are those for $\mu_R = 1$ GeV. In the normalisation of Eq. (21), the remainders are therefore

$$\begin{aligned}\mathcal{M}_i^{(1)} &= \mathcal{M}_{i,\text{publ}}^{(1)} + 2\beta_0 \ln \frac{\mu_R^2}{\text{GeV}^2}\, \mathcal{M}_i^{(0)},\\ \mathcal{M}_i^{(2)} &= \frac{1}{64}\, \mathcal{M}_{i,\text{publ}}^{(2)},\end{aligned} \tag{24}$$

with $\beta_0 = (11N_c - 2n_f)/3$ and masses in GeV. The added term restores the scale dependence required by the renormalisation group equations; with both conversions in place, the released one- and two-loop remainders are consistent with each other at every tested point. Appendix B gives the evidence. The conversion leaves the powers of the threshold logarithms and the relative truncation errors unchanged. The numerical two-loop coefficients, and their comparison with lower orders, require the common normalisation; restoring the one-loop scale dependence changes the one-loop constant and the finite virtual corrections of Sec. V.

The one-loop amplitude with one O_G and one O_J insertion enters at $\mathcal{O}(a_s^4)$, the order of the two-loop O_G–O_G amplitude, since $C_J = \mathcal{O}(a_s)\, C_G$. It is proportional to the tree amplitude: in the normalisation of Eqs. (3.6) and (3.7) of Ref. [7],

$$\mathcal{M}_{i,GJ,\text{publ}}^{(1),\text{fin}} = -48\, n_f\, \mathcal{M}_i^{(0)}, \qquad i = 1, 2, \tag{25}$$

at every point of the physical phase space. The proportionality is consistent with the axial anomaly, which relates the divergence of the flavour singlet axial current to the gluonic pseudoscalar operator [13, 15, 33]. At $\mathcal{O}(a_s^4)$ the mixed contribution therefore has the kinematic dependence of the tree amplitude and adds no logarithm of p_T.

E. Perturbative order

In Eq. (21), the superscript (n) counts loops of gluons and light quarks in the effective theory; in a two-Higgs-doublet model the top quark loops are contained in the Wilson coefficients. The perturbative order of a contribution to the cross section also depends on the powers of a_s in the Wilson coefficients.

Since $C_G \propto a_s$, the Type II tree, one-loop and two-loop contributions to the full amplitude are of order a_s^2, a_s^3 and a_s^4, respectively. In a 2HDM, the leading amplitude is of Type I and starts at $\mathcal{O}(a_s)$, giving a LO cross section of order a_s^2. Interference between the leading Type I amplitude and the Type II tree amplitude contributes at order a_s^3, that is at NLO. This is the one-particle reducible contribution to the virtual corrections in Refs. [1, 5]. The Type II two-loop amplitude first enters through its interference with the leading Type I amplitude at order a_s^5, or N^{3}LO. The same counting applies to the corresponding class of amplitudes for Higgs pairs [9]. The mixed O_G–O_J contribution of Sec. II D, of the same order a_s^4 as the two-loop O_G–O_G amplitude, also contributes at this N^{3}LO.

When no Type I amplitude is present, the Type II tree amplitude instead gives the LO cross section. This is the case for an axion-like particle whose only relevant interaction is its coupling to gluons through $G\tilde{G}$. The one-loop and two-loop terms in Eq. (21) then provide virtual contributions at NLO and NNLO, respectively. The present results therefore supply the O_G–O_G part of the two-loop virtual correction at NNLO for this process. A contact interaction of dimension six containing two pseudoscalars and two gluon fields would introduce an additional Type I amplitude, as in Ref. [11].

F. Finite top quark mass effects

The threshold expansion and the heavy top quark approximation involve different limits. Taking $\beta_A \to 0$ places the pair invariant mass close to $2m_A$, but imposes no restriction on m_A/m_t. Finite top quark mass effects must therefore be assessed separately when A is the pseudoscalar of a 2HDM, at values of $\tan\beta$ for which the top quark loop dominates. No heavy top quark approximation is involved when the coupling to $G\tilde{G}$ is specified directly as an effective interaction, as in the axion-like particle case above.

For the Type II tree amplitude, the dependence on the top quark mass is known. Each triangle in the full theory couples a pseudoscalar to one on-shell gluon and one virtual gluon. The two triangles in a given channel have the same form factor, so the contribution of each channel is multiplied, relative to its heavy top quark limit, by

$$\left[\frac{I_2(\tau_A, \lambda)}{I_2^\infty}\right]^2, \quad \tau_A = \frac{4m_t^2}{m_A^2}, \quad \lambda = \frac{4m_t^2}{t} \quad \text{or} \quad \frac{4m_t^2}{u}. \tag{26}$$

Here I_2 is the loop function that also appears in $A \to Z\gamma$ [1], with λ evaluated at the virtuality of the exchanged gluon in the t and u channel, respectively; I_2^∞ denotes its heavy top quark limit. At the pair threshold, $t = u = -m_A^2$, so both channels have the same factor: 1.006, 1.013 and 1.022 for $m_A = 125$, 150 and 170 GeV, respectively, with $m_t = 172.5$ GeV. These differ from unity by at most about 2.2%.

For the loop corrections, a partial estimate can be made for the one-loop DPS logarithm. Its coefficient involves the amplitudes for $gg \to A$ with both gluons on shell, as discussed in Sec. IV C. The relevant triangle form factor is therefore obtained in the limit $\lambda \to \infty$, rather than at the spacelike virtuality of the exchanged gluon

in the tree diagrams:

$$\frac{I_2(\tau_A,\infty)}{I_2^{\infty}} = \tau_A f(\tau_A), \qquad f(\tau) = \arcsin^2\!\left(\frac{1}{\sqrt{\tau}}\right). \quad (27)$$

This form holds for $\tau_A \geq 1$, which includes the masses considered here. If the DPS factorisation applies to the full theory, with the top quark triangles treated as hard subgraphs, the coefficient of the one-loop logarithm is multiplied by $[\tau_A f(\tau_A)]^2$, one factor for each of the two subprocesses $gg \to A$; for these masses, this is also the ratio of the full to the heavy top quark LO cross section for $gg \to A$. This gives 1.096, 1.145 and 1.196 for the same three masses, while the velocity dependence and the helicity selection rule remain unchanged. Neither this estimate nor the comparison of tree amplitudes determines the finite top quark mass corrections to the remaining one-loop terms or to the complete two-loop amplitude.

III. THRESHOLD EXPANSION

The calculation has four steps, each checked independently. First, the finite remainders of Ref. [7] are split into their flavour coefficients and brought to the normalisation of Eq. (24) (App. B). Second, each flavour coefficient is expanded about $\beta_A = 0$ at fixed angle, through β_A^4 and, independently, through β_A^6 (Sec. III A). Third, the constants that arise at the endpoints are reduced to a basis of eight transcendental constants, after which the inverse and odd powers of β_A cancel identically (Sec. III B and App. A). Fourth, the series are compared with the unexpanded remainders, evaluated in Mathematica at increasing working precision and, at selected points, with an independent Python code (Sec. III C). The leading power terms are analysed in Sec. IV, where forward limits of the one-loop remainders and a separate one-loop $gg \to HH$ calculation test their relation to the DPS singularity. Their consequences for the finite virtual contribution are discussed in Sec. V. The software and the computing environment are described in App. D.

A. Construction

The remainders of Ref. [7] are functions of x, y and z (Eq. (10)). We replace these variables by their expressions in β_A and c from Eq. (12), and expand each flavour coefficient about $\beta_A = 0$ at fixed c,

$$\mathcal{M}^{(2)}_{i,r}(\beta_A, c) = \sum_{n=-4}^{p} \sum_{k} C^{(i,r)}_{nk}\, \beta_A^n \ln^k \beta_A + \mathcal{O}_{\log}(\beta_A^{p+1}), \quad (28)$$

with $r = 0, 1, 2$ and $p = 4$ or 6. The coefficients $C^{(i,r)}_{nk}$ depend on c, m_A^2, μ_R^2 and N_c, and $\mathcal{O}_{\log}(\beta_A^n)$ stands for β_A^n times a polynomial in $\ln\beta_A$. Inverse powers appear in intermediate steps because the tensor basis and the rational prefactors contain inverse Gram determinants, Eq. (13). They must be kept until all contributions at a given power have been combined.

The remainders contain logarithms, classical polylogarithms and the function $\mathrm{Li}_{2,2}$ of weight four [34, 35], whose arguments approach correlated endpoints as $\beta_A \to 0$. We expand the rational prefactors exactly and obtain the expansions of the special functions from differential equations in β_A, with boundary values at $\beta_A = 0$. These equations have the iterated-integral form familiar from harmonic polylogarithms [36] and from canonical differential equations for Feynman integrals [37]; the map $x = (1-\beta_A)/(1+\beta_A)$ is one of the transformations under which harmonic polylogarithms are closed [36]. The method thus expands the known remainders, not the loop integrals, which could instead be expanded near threshold by the method of regions [38]. Expansions of two-loop gluon fusion amplitudes in kinematic limits are also used to obtain compact representations that are fast to evaluate, for example for small transverse momentum and at high energy [39, 40] and about the top quark pair threshold [41]; the present expansion is about the pair production threshold of the massless remainders. For the dilogarithm, for example, $\mathrm{d}\,\mathrm{Li}_2(x)/\mathrm{d}\beta_A = 2(1-\beta_A^2)^{-1}\ln[2\beta_A/(1+\beta_A)]$ with $\mathrm{Li}_2(1) = \pi^2/6$ gives

$$\mathrm{Li}_2(x) = \frac{\pi^2}{6} + 2\beta_A\big[\ln(2\beta_A) - 1\big] - \beta_A^2 + \beta_A^3\left[\frac{2}{3}\ln(2\beta_A) + \frac{1}{9}\right] - \frac{2}{3}\beta_A^4 + \mathcal{O}\big(\beta_A^5 \ln\beta_A\big). \quad (29)$$

Since the prefactors start at β_A^{-4}, the special functions have to be expanded four orders beyond the final accuracy. The fourth and sixth order expansions were constructed independently, and all coefficients common to both agree.

B. Endpoint constants and exact cancellations

The boundary values at $\beta_A = 0$ introduce thirteen constants $\mathrm{MB}[p, q, a, w]$, defined in Eq. (A1) of App. A. Their arguments take the values ± 1 and $\frac{1}{2}$; the integer relation search below reduces them to a basis of constants of weight three and four, of the type that occurs for alternating Euler sums [42]. An integer relation search with the PSLQ algorithm [43] at 110 digits expresses each of them in the basis

$$\{\zeta_3,\ \pi^2\ln 2,\ \ln^3 2,\ \pi^4,\ \pi^2\ln^2 2,\ \ln^4 2,\ \mathrm{Li}_4(\tfrac{1}{2}),\ \zeta_3\ln 2\}, \quad (30)$$

with a largest residual of 4×10^{-107} (App. A).

With these reductions, and with the dilogarithms of $(1-c)/2$, $(c-1)/(c+1)$ and $(c+1)/(c-1)$ rewritten through $\mathrm{Li}_2\big((1+c)/2\big)$ by the reflection and Landen identities, after extracting the overall factor m_A^2, the co-

efficient of each power β_A^n is a polynomial in

$$\{\ln\beta_A,\ \ln(1\pm c),\ \mathrm{Li}_2\big(\tfrac{1+c}{2}\big),\ \ln\tfrac{m_A^2}{\mu_R^2}\}\cup\{\ln 2,\ \pi,\ \zeta_3,\ \mathrm{Li}_4(\tfrac{1}{2})\} \tag{31}$$

with coefficients rational in c and N_c. Treating these quantities as algebraically independent, we find that

1. the coefficients of $\beta_A^{-4},\dots,\beta_A^{-1}$ vanish identically in every flavour coefficient;
2. the coefficients of β_A, β_A^3 and β_A^5 vanish identically, so that through β_A^6 the expansion is a series in β_A^2 with logarithms;
3. the coefficients of β_A^0, β_A^2, β_A^4 and β_A^6 of all six flavour coefficients, 24 in total, are even in c, so that Bose symmetry holds order by order.

The cancellations and the reductions of the constants were confirmed independently in Mathematica.

The cancellations have a simple origin. Exchanging the pseudoscalars exchanges t and u, so the remainders depend on the kinematics through invariants that are symmetric under this exchange,

$$\hat{s}=\frac{4m_A^2}{1-\beta_A^2},\qquad (t-u)^2=\hat{s}^2\beta_A^2c^2,\qquad tu-m_A^4=\hat{s}\,p_T^2, \tag{32}$$

with p_T^2 given in Eq. (13). The difference $t-u=\hat{s}\beta_A c$, which is odd in β_A, enters only through its square. Each invariant depends on β_A only through β_A^2 and on c only through c^2, so a function analytic in them at threshold is a series in β_A^2 and even in c. Odd or inverse powers of β_A would therefore signal non-analytic behaviour at threshold. There is no threshold for a pair of massive internal particles at $\hat{s}=4m_A^2$, since the loops contain no massive particles, and the colourless final state has no QCD Coulomb singularity. These observations alone do not exclude other Landau singularities: the singularity at $p_T\to 0$ discussed in Sec. IV C is one, and at one loop it is logarithmic [2]. The Gram determinant $tu-m_A^4$ vanishes at the same point, but its inverse powers come from the tensor basis and the rational prefactors; they are not singularities of the amplitude. In the present calculation, the explicit expansion shows that the inverse powers cancel and that the remaining singular behaviour is logarithmic in p_T. A remainder of the form $\sum_k a_k \ln^k p_T^2$, with coefficients a_k analytic in the invariants of Eq. (32) at threshold, is therefore a series in β_A^2 with logarithms and has no inverse powers. This accounts for the form of the cancellations found above. At two loops it is not an independent proof, because there the logarithmic nature of the singularity is established only by the calculation itself.

Table I lists the leading terms. Since the odd powers vanish, the first omitted term is expected to be $\mathcal{O}_{\log}(\beta_A^{p+2})$ rather than the $\mathcal{O}_{\log}(\beta_A^{p+1})$ of Eq. (28). The truncation errors should therefore scale as β_A^6 for $p=4$ and as β_A^8 for $p=6$, up to logarithmic factors.

TABLE I. Leading terms of the threshold expansion of the two-loop Type II remainders. The second column lists the inverse powers with nonzero coefficients in the construction files; they cancel identically after the reductions of Sec. III B, as do all odd powers. The last column gives the powers of $\mathcal{L}$, Eq. (36), at leading power.

quantity	inverse powers in construction	leading power	powers of $\mathcal{L}$
$\mathcal{M}^{(2)}_{1,0}$	β_A^{-2}, β_A^{-1}	β_A^0	0, 1, 2, 3
$\mathcal{M}^{(2)}_{1,1}$	–	β_A^0	0, 1, 2
$\mathcal{M}^{(2)}_{1,2}$	–	β_A^0	0
$\mathcal{M}^{(2)}_{2,0}$	β_A^{-4}, β_A^{-3}, β_A^{-2}, β_A^{-1}	β_A^0	0, 1, 2
$\mathcal{M}^{(2)}_{2,1}$	β_A^{-1}	β_A^0	0, 1
$\mathcal{M}^{(2)}_{2,2}$	–	β_A^2	0

The coefficient of n_f^2 is the simplest:

$$\mathcal{M}^{(2)}_{1,2}(\beta_A,c)=K+\mathcal{O}(\beta_A^2), \tag{33}$$

$$\mathcal{M}^{(2)}_{2,2}(\beta_A,c)=\beta_A^2(1-c^2)K+\mathcal{O}(\beta_A^4), \tag{34}$$

where K does not depend on the angle. It is the only flavour coefficient that follows the tree-level pattern $\mathcal{M}_2\propto\beta_A^2(1-c^2)$ of Eq. (20), with the same ratio to the tree amplitude in both tensor structures. The complete expressions through β_A^6 are released as Wolfram Language files (App. D).

C. Accuracy

We compared the series with the unexpanded remainders on the grid in Table II, which contains 9 values of β_A between 10^{-3} and 3/10, 5 scattering angles and 3 combinations of mass, renormalisation scale and number of colours. At each of the 135 points, the comparison included the six flavour coefficients and the two complete coefficients, giving 1080 comparisons in total. Each unexpanded remainder was evaluated with 80, 120, 160, 220 and 300 digits of working precision, and a value was accepted once two successive precisions agreed to a relative difference below 10^{-25}, with at least 30 significant digits in both. The parameter set with $N_c\neq 3$ was included to test the colour dependence; all physical results use $N_c=3$. The relative truncation error is defined as

$$E_i^{(p)}=\frac{\left|\mathcal{M}_i^{[p]}-\mathcal{M}_i\right|}{|\mathcal{M}_i|},\qquad p=4,6, \tag{35}$$

where $\mathcal{M}_i$ is the unexpanded remainder and $\mathcal{M}_i^{[p]}$ its series through β_A^p.

Figure 2 shows these errors for the complete coefficients at $n_f=5$. The slopes of the curves, discussed below, test the scaling β_A^{p+2} expected from the cancellation of the odd powers. Because each curve is a maximum over

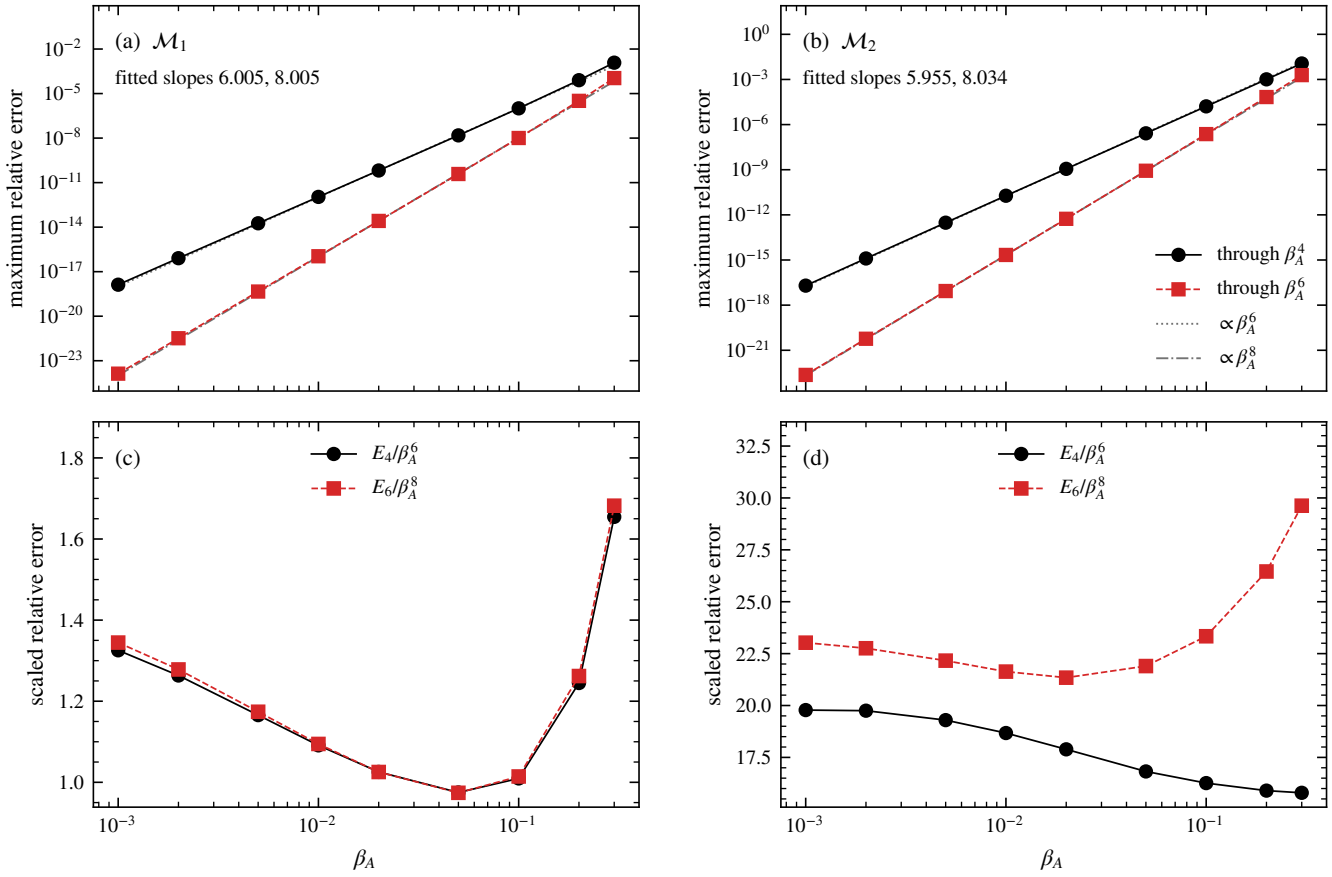


FIG. 2. Relative truncation errors, Eq. (35), of the complete two-loop coefficients at $n_f = 5$: $\mathcal{M}_1$ (left) and $\mathcal{M}_2$ (right). Upper panels: maximum over the angles and parameter sets of Table II for the series through β_A^4 (circles) and through β_A^6 (squares), with lines proportional to β_A^6 and β_A^8. Lower panels: the same maxima divided by β_A^6 and β_A^8.

TABLE II. Validation grid. At each point the six flavour coefficients and the two complete $n_f = 5$ coefficients are compared.

quantity	values
β_A	1/1000, 1/500, 1/200, 1/100, 1/50, 1/20, 1/10, 1/5, 3/10
c	$-2/3$, $-1/3$, 0, 1/3, 2/3
$(m_A/\mathrm{GeV},\ \mu_R/\mathrm{GeV},\ N_c)$	$(125, 125, 3)$, $(125, 250, 3)$, $(150, 75, 4)$

TABLE III. Maximum relative errors of the complete Type II coefficients on the validation grid. At each value of β_A, the maximum is taken over the 5 angles and 3 parameter sets of Table II.

	$\mathcal{M}_1$ through		$\mathcal{M}_2$ through	
β_A	β_A^4	β_A^6	β_A^4	β_A^6
0.01	1.09×10^{-12}	1.09×10^{-16}	1.87×10^{-11}	2.16×10^{-15}
0.05	1.52×10^{-8}	3.81×10^{-11}	2.63×10^{-7}	8.55×10^{-10}
0.10	1.01×10^{-6}	1.01×10^{-8}	1.63×10^{-5}	2.33×10^{-7}
0.20	7.97×10^{-5}	3.23×10^{-6}	1.02×10^{-3}	6.77×10^{-5}
0.30	1.21×10^{-3}	1.10×10^{-4}	1.15×10^{-2}	1.94×10^{-3}

the points of Table II, a change of the point that gives the maximum as β_A varies would distort these slopes. For the series through β_A^6 this does not occur: for both coefficients and at every velocity the maximum comes from the same point, $c = 0$ with $m_A = 150$ GeV, $\mu_R = 75$ GeV and $N_c = 4$, so the curves follow the error of a single point. For the series through β_A^4 the same holds for $\mathcal{M}_1$, while for $\mathcal{M}_2$ the maximum alternates between two parameter sets at $|c| = 2/3$. The maximum is also representative of the grid: at each velocity and for both orders the errors at all other points lie within a factor of eight of it.

For the series through β_A^4, the fitted slopes of the logarithm of the maximum error against $\ln \beta_A$ over the plotted range are 6.005 for $\mathcal{M}_1$ and 5.955 for $\mathcal{M}_2$; for the series through β_A^6 they are 8.005 and 8.034. The expected values are 6 and 8 (Sec. III B), while a surviving odd power would give values close to 5 and 7. The small

TABLE IV. Relative truncation errors $E_i^{(6)}$ of the complete $n_f = 5$ coefficients towards the beam, at $m_A = \mu_R = 125\,\mathrm{GeV}$ and $N_c = 3$. The last rows give the largest value on the grid of Table II. The values at $c = -0.99$ equal those at $-c$.

	c	$\beta_A = 0.1$	$\beta_A = 0.2$	$\beta_A = 0.3$
$\mathcal{M}_1$	0.9	6.36×10^{-9}	1.11×10^{-6}	1.78×10^{-5}
$\mathcal{M}_1$	0.99	1.06×10^{-8}	2.43×10^{-6}	5.76×10^{-5}
$\mathcal{M}_1$	0.999	1.23×10^{-8}	2.96×10^{-6}	7.18×10^{-5}
$\mathcal{M}_2$	0.9	1.39×10^{-7}	3.08×10^{-5}	6.68×10^{-4}
$\mathcal{M}_2$	0.99	6.68×10^{-8}	1.47×10^{-5}	2.88×10^{-4}
$\mathcal{M}_2$	0.999	6.21×10^{-8}	1.65×10^{-5}	3.74×10^{-4}
$\mathcal{M}_1$	grid max.	1.01×10^{-8}	3.23×10^{-6}	1.10×10^{-4}
$\mathcal{M}_2$	grid max.	2.33×10^{-7}	6.77×10^{-5}	1.94×10^{-3}

deviations from 6 and 8 are consistent with the powers of $\ln \beta_A$ and the higher terms omitted from the series.

In the lower panels, exact power behaviour would give horizontal lines. The scaled errors vary slowly, because both the omitted terms and the remainders that normalise the errors contain powers of $\ln \beta_A$, and over the tested range they change by less than a factor of two. An incorrect coefficient at a retained order would instead make them grow as an inverse power of β_A as β_A decreases. Towards $\beta_A = 0.3$ the scaled errors of the series through β_A^6 rise for both coefficients, as the terms beyond the first omitted order begin to contribute.

Table III lists the maximum errors at selected velocities. For $\beta_A \le 0.2$, the series through β_A^6 reproduces $\mathcal{M}_1$ and $\mathcal{M}_2$ with relative errors of at most 3.23×10^{-6} and 6.77×10^{-5}; at $\beta_A = 0.3$ the errors are 1.10×10^{-4} and 1.94×10^{-3}. Bose symmetry was checked by comparing the values at c and $-c$: they are identical in all stored digits in 432 comparisons of the unexpanded coefficients and 864 of the series. The reference values were computed in Mathematica from the expressions of Ref. [7]. To check their evaluation independently, We wrote a separate Python code with mpmath [44]. It reads the published files with its own parser and computes $\mathrm{Li}_{2,2}$ from its defining double sum, or from a one-dimensional integral representation where the sum converges slowly. At four points covering the three parameter sets, the two complete coefficients agree with the reference values to a relative difference of at most 1.3×10^{-40}, the precision to which the reference values are stored. The six flavour coefficients, compared at $\beta_A = 1/10$ and $c = 1/3$, agree to 1.6×10^{-40}. At $c = 0$, where $t = u$, the published expressions have removable singularities and are evaluated as limits.

The expansion is made at fixed angle, but p_T vanishes as $c \to \pm 1$ at fixed β_A, so the series must also be tested close to the beam axis. We therefore compared it with the unexpanded remainders at $c = 0.9$, 0.99, 0.999 and -0.99 for $\beta_A = 0.1$, 0.2 and 0.3. Table IV gives the relative errors of the series through β_A^6. All values are below the maximum on the main grid at the same β_A, except for $\mathcal{M}_1$ at $\beta_A = 0.1$, where the largest value, 1.23×10^{-8} at $c = 0.999$, exceeds it by 21%. The series thus remains accurate close to the beam, consistent with its coefficients having no poles at $c = \pm 1$, only logarithms of $1 \pm c$ (Sec. IV A).

IV. LEADING POWER AND THE DOUBLE PARTON SCATTERING SINGULARITY

A. Leading power

To determine the leading threshold behaviour of the two-loop remainders, we first collect the terms of order β_A^0, including all powers of $\ln \beta_A$ that multiply them. The notation $\big|_{\beta_A^0}$ used below denotes this part of the expansion.

In the expanded remainders, the logarithms of β_A and of the angular variables appear at leading power in the combination $\ln \beta_A + \frac{1}{2} \ln(1 - c^2)$. We therefore introduce

$$\begin{aligned} \mathcal{L} &= \ln \beta_A + \tfrac{1}{2} \ln(1 - c^2) + \ln 2 + \frac{i\pi}{2} \\ &= \tfrac{1}{2} \ln\bigl[-4\beta_A^2 (1 - c^2)\bigr], \end{aligned} \tag{36}$$

where the logarithm of the negative argument is taken on the branch $\ln(-X) = \ln X + i\pi$ for $X > 0$, which is the one followed by the remainders in the physical region.

The meaning of $\mathcal{L}$ follows from Eq. (13). Substituting the expressions for $tu - m_A^4$ and p_T^2 gives

$$\begin{aligned} \mathcal{L} &= \tfrac{1}{2} \ln \frac{m_A^4 - tu}{m_A^4} + \ln(1 - \beta_A^2) \\ &= \ln \frac{2 p_T}{m_A} + \frac{i\pi}{2} + \tfrac{1}{2} \ln(1 - \beta_A^2). \end{aligned} \tag{37}$$

The last term is of order β_A^2. Near threshold, $\mathcal{L}$ is therefore the logarithm of the transverse momentum of the pseudoscalars, up to the constant phase $i\pi/2$ and power corrections.

When the leading power terms are rewritten in terms of $\mathcal{L}$, no separate $\ln(1 \pm c)$ or $\mathrm{Li}_2\bigl((1+c)/2\bigr)$ remains, and the rational dependence on c cancels. All dependence on the scattering angle is then contained in $\mathcal{L}$, and the coefficients of its powers are independent of c. For general N_c and n_f,

$$\mathcal{M}_1^{(2)}\big|_{\beta_A^0} = m_A^2 \sum_{k=0}^{3} \widehat{A}_{1k}\, \mathcal{L}^k, \tag{38}$$

$$\mathcal{M}_2^{(2)}\big|_{\beta_A^0} = m_A^2 \sum_{k=0}^{2} \widehat{A}_{2k}\, \mathcal{L}^k. \tag{39}$$

The highest logarithmic coefficients are

$$\begin{aligned} \widehat{A}_{13} &= -\tfrac{28}{3} \pi N_c^2, & \widehat{A}_{12} &= -\tfrac{14}{3} \pi N_c (11 N_c - 2 n_f), \\ \widehat{A}_{22} &= -2\pi N_c^2, & \widehat{A}_{21} &= -\tfrac{4}{3} \pi N_c (4 N_c - n_f), \end{aligned} \tag{40}$$

TABLE V. Coefficients of the leading power term, $\mathcal{M}_i|_{\beta_A^0} = m_A^2 \sum_k \widehat{A}_{ik}\,\mathcal{L}^k$, with $\mathcal{L}$ of Eq. (36). They do not depend on the scattering angle. Exact forms for $\mu_R = m_A$ and general N_c, n_f; numerical values for $N_c = 3$, $n_f = 5$, $\mu_R = m_A$.

	k	exact	$\mathrm{Re}\,\widehat{A}_{ik}$	$\mathrm{Im}\,\widehat{A}_{ik}$
$\mathcal{M}_1$	3	$-\frac{28}{3}\pi N_c^2$	-2.63894×10^2	0
$\mathcal{M}_1$	2	$-\frac{14}{3}\pi N_c(11N_c - 2n_f)$	-1.01159×10^3	0
$\mathcal{M}_1$	1	Eq. (41)	3.55129×10^3	4.76702×10^3
$\mathcal{M}_1$	0	–	-3.98215×10^2	-4.00569×10^3
$\mathcal{M}_2$	2	$-2\pi N_c^2$	-5.65487×10^1	0
$\mathcal{M}_2$	1	$-\frac{4}{3}\pi N_c(4N_c - n_f)$	-8.79646×10^1	0
$\mathcal{M}_2$	0	–	3.12007×10^2	3.40501×10^2

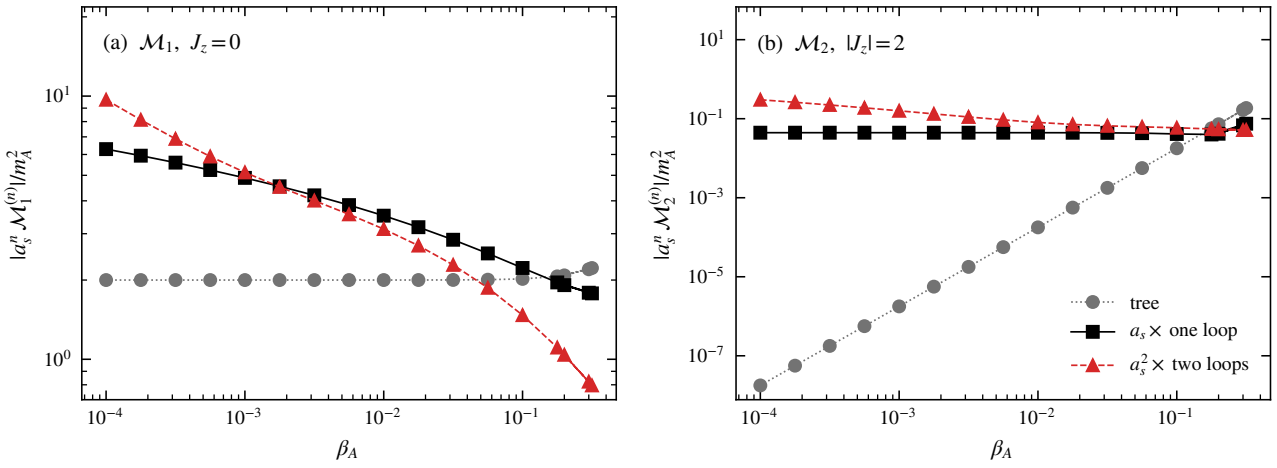


FIG. 3. Tree-level, one-loop and two-loop terms of the Type II amplitude, $|a_s^n \mathcal{M}_i^{(n)}|/m_A^2$ with $a_s = 0.0094$ ($\alpha_s = 0.118$), at $c = 1/3$, $m_A = \mu_R = 125\,\mathrm{GeV}$, $N_c = 3$ and $n_f = 5$. Tree level: Eq. (20); one loop: remainder of Ref. [7] with the scale dependence of Eq. (24) restored; two loops: series through β_A^6. $\mathcal{M}_1$ and $\mathcal{M}_2$ are the equal and opposite helicity amplitudes, Eq. (18).

which do not depend on μ_R, and

$$\widehat{A}_{11} = \frac{\pi N_c}{9}\big(N_c\, a_\mathrm{A} - n_f\, a_\mathrm{F}\big), \tag{41}$$

where

$$a_\mathrm{A} = 1454 + 462\ln 2 + 252\ln^2 2 - 42\pi^2 + 693\,i\pi - 693 L_\mu,$$
$$a_\mathrm{F} = 152 + 84\ln 2 + 126\,i\pi - 126 L_\mu,$$

and $L_\mu = \ln(m_A^2/\mu_R^2)$. Table V gives all coefficients for $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$. At β_A^2, β_A^4 and β_A^6 the same substitution again removes every $\ln(1 \pm c)$ and Li_2; the remaining dependence on the angle is rational, even in c and free of poles at $c = \pm 1$.

Up to the constant $i\pi/2$ and power corrections, $\mathcal{L}$ is the logarithm of the transverse momentum of the pseudoscalars or, equivalently, of the Gram determinant $tu - m_A^4$. The exact one-loop remainders already contain it, since they depend on

$$\ln(y - x) + \ln(z - x) \text{ and}$$
$$(y - x)(z - x) = yz - 1 = (tu - m_A^4)/m_A^4$$

by Eq. (10). The two-loop coefficients have no finite limit as $\beta_A \to 0$; they grow as $\mathcal{L}^3$ and $\mathcal{L}^2$ (Fig. 3).

B. One loop

Fitting the one-loop remainders of Ref. [7] at $\beta_A = 10^{-40}$ and 10^{-50} with 400-digit arithmetic, for $N_c = 3$, 4 and 5, $n_f = 0$, 2 and 5, two values of μ_R and several angles, we find to 30 digits

$$\mathcal{M}_1^{(1)} = m_A^2\big[7\pi N_c\,\mathcal{L} + \widehat{A}_{10}^{(1)}\big] + \mathcal{O}(\beta_A),$$
$$\mathcal{M}_2^{(1)} = \tfrac{1}{2}\pi N_c\, m_A^2 + \mathcal{O}(\beta_A), \tag{42}$$

with $\widehat{A}_{10}^{(1)} = -93.900 + 11.484\,i$ for $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$, in the normalisation of Eq. (24). The constant does not depend on the angle, and by Eq. (24), its scale dependence is $\partial \widehat{A}_{10}^{(1)}/\partial \ln \mu_R^2 = 4i\beta_0$. The coefficient of $\mathcal{L}$ and the limit of $\mathcal{M}_2^{(1)}$ depend neither on n_f nor on μ_R. Together with Eqs. (20) and (39), the behaviour at threshold is summarised in Table VI. The $|J_z| = 2$ amplitude is suppressed near threshold only at tree level.

TABLE VI. Behaviour of the Type II coefficients at threshold, $\beta_A \to 0$ at fixed angle: tree-level values, and the highest power of $\mathcal{L}$ at one and two loops.

	tree level	one loop	two loops
$\mathcal{M}_1$ $(J_z = 0)$	$2im_A^2$	$\mathcal{L}$	$\mathcal{L}^3$
$\mathcal{M}_2$ $(\lvert J_z\rvert = 2)$	$2im_A^2\beta_A^2(1-c^2)$	$\mathcal{L}^0$	$\mathcal{L}^2$

According to Eq. (18), $\mathcal{A}_{\pm\mp}$ retains the azimuthal factor $e^{\pm 2i\phi}$ as $|\vec{p}_3| \to 0$. The loop amplitude therefore has a direction-dependent limit and is not analytic in $\vec{p}_3$ at the origin. At one loop $\mathcal{M}_2$ approaches a nonzero constant without a DPS logarithm; at two loops it also acquires the logarithm discussed in Sec. IV D.

C. The double parton scattering singularity

a. Origin of the logarithm. The logarithm $\mathcal{L}$ in Eq. (42) can be understood from the one-loop Type II diagrams near $p_T = 0$, shown in Fig. 1(b). As $p_T \to 0$, all four internal gluons can be on shell simultaneously, two of them collinear to each incoming gluon. This configuration is the pinch that produces the DPS singularity of the crossed box [2]. The singular contribution can be expressed as an integral over transverse momentum of the product of the light cone wavefunctions of the two incoming gluons and the tree amplitudes of the two hard subprocesses [2]. The wavefunctions describe the splitting of each incoming parton. The hard subprocess amplitudes are evaluated with collinear on-shell incoming partons and zero transverse momentum. Thus, the splittings determine the strength of the singularity, while the hard subprocesses enter its coefficient. In the present case, the splittings are $g \to gg$ and the subprocesses are $gg \to A$ through O_G.

For a gluon splitting into a collinear pair, the daughter helicities add to 0 or ± 2, whereas the parent helicity is ± 1. The numerator of the splitting vertex therefore vanishes in the exactly collinear limit; for the allowed tree helicity configurations it is linear in the relative transverse momentum. After including the propagator, the corresponding light cone wavefunction behaves as the inverse of that momentum. The transverse momentum integral is consequently at most logarithmically singular, as in the massless quark examples of Ref. [2].

b. Helicity selection and weights. The logarithm is present only for certain helicities of the partons in the singular configuration. For the fermion loops of Ref. [2], helicity conservation forces the four intermediate partons into $J_z = 0$; for gluon loops only conservation of J_z between the incoming state and the intermediate partons remains [3]. Here the effective vertices restore the restriction. The amplitude for $gg \to A$ vanishes unless the two gluons have equal helicities, and the gluons of each fusing pair move in opposite beam directions, so each pair has $J_z = 0$. A DPS logarithm is therefore possible only for incoming gluons with $J_z = 0$, which is why $\mathcal{M}_1$ has a logarithm at one loop and $\mathcal{M}_2$ does not.

For a positive-helicity gluon splitting into gluons a and b with longitudinal momentum fractions z and $1-z$, the longitudinal weights, up to coupling, colour and transverse momentum factors, are

$$\begin{aligned} w_{++}(z) &= \frac{1}{\sqrt{z(1-z)}}, \\ w_{+-}(z) &= \frac{z^2}{\sqrt{z(1-z)}}, \\ w_{-+}(z) &= \frac{(1-z)^2}{\sqrt{z(1-z)}}, \end{aligned} \tag{43}$$

where the subscripts give (h_a, h_b) [45, 46]. The sum of their squares, $[1+z^4+(1-z)^4]/[z(1-z)]$, is the z dependence of the $g \to gg$ splitting function. At the DPS point, the first pseudoscalar receives a fraction x of the first incoming momentum and, since the two pseudoscalars have equal masses, a fraction $1-x$ of the second [2]; in Fig. 1(b), $k = xp_1$ and $p_3 - k = (1-x)p_2$. Here x and z are momentum fractions, not the variables of Eq. (10).

Since A is odd under parity, which reverses all helicities, the amplitudes for $g^+g^+ \to A$ and $g^-g^- \to A$ have opposite signs [46]. For two incoming positive-helicity gluons, with the first helicity of each daughter pair referring to the gluon that goes to the first pseudoscalar, each vertex requires equal helicities, so both splittings produce the same pair: $(+,+)$, $(+,-)$ or $(-,+)$. In the mixed cases one vertex receives two negative-helicity gluons, and these contributions enter with the opposite sign. The factorised form of Ref. [2] then gives

$$\begin{aligned} F_A(x) &= w_{++}(x)w_{++}(1-x) \\ &\quad - w_{+-}(x)w_{+-}(1-x) - w_{-+}(x)w_{-+}(1-x) \\ &= \frac{1}{x(1-x)} - 2x(1-x), \end{aligned} \tag{44}$$

since each mixed product equals $x(1-x)$. For a CP-even coupling to G^2 the two signs are equal and the mixed terms add:

$$F_S(x) = \frac{1}{x(1-x)} + 2x(1-x). \tag{45}$$

These weights differ from those of the massless quark boxes for $gg \to HH$ in Ref. [2], where internal quarks of equal helicity carry complementary momentum fractions and the weight is proportional to $x(1-x)$.

c. Normalisation and the forward limit. At threshold, both pseudoscalars are at rest and $x = \frac{1}{2}$, so that $F_A(\frac{1}{2}) = \frac{7}{2}$ and $F_S(\frac{1}{2}) = \frac{9}{2}$. By Eq. (36), the coefficient of $\ln p_T^2$ is half the coefficient of $\mathcal{L}$. The one-loop threshold result in Eq. (42) therefore gives

$$\frac{7\pi N_c\, m_A^2}{2} = \frac{7}{2}\pi N_c m_A^2 = \pi N_c m_A^2 F_A\left(\frac{1}{2}\right).$$

Matching at threshold thus fixes the overall factor multiplying F_A to $\pi N_c m_A^2$. This factor is obtained from the calculated amplitude; it has not been derived independently from the factorisation formula.

Having fixed this constant, the predicted dependence on x can be tested away from threshold. The forward limit $c \to 1$ at fixed velocity also gives $p_T \to 0$, but the produced bosons now move along the beam. Since the test is also applied to $gg \to HH$ later, we denote the velocity and mass of the produced bosons by β and m, with $\beta = \beta_A$ and $m = m_A$ for AA. In this limit, the longitudinal velocity of the first boson is $2x - 1$, so $x = (1+\beta)/2$.

To extract the logarithmic coefficient, I differentiate the real part of the one-loop remainder with respect to $\ln p_T^2$ at fixed β and m, and then take the forward limit. Dividing by the normalisation fixed above defines

$$D(\beta) \equiv \frac{1}{\pi N_c m^2} \lim_{c\to 1} \frac{\partial\, \mathrm{Re}\, \mathcal{M}_1^{(1)}}{\partial \ln p_T^2} = F_A\left(\frac{1+\beta}{2}\right) \qquad (gg \to AA). \tag{46}$$

The last equality is the prediction to be tested: no further adjustment of the normalisation is made as β is varied. We evaluated $D(\beta)$ from the exact one-loop remainders using $1 - c = 10^{-14}$ and 10^{-18}, over $0.001 \le \beta \le 0.7$, with $N_c = 3$ and 4 and $n_f = 0$, 2 and 5. The largest relative deviation from Eq. (46) was 3×10^{-15}.

d. Comparison with scalar pair production. The scalar process provides a separate test of the relative signs in the helicity sum. The splittings and colour factors are unchanged, while replacing the CP-odd vertices by CP-even ones changes F_A to F_S. We tested this prediction using the one-loop amplitudes for $gg \to HH$ in which each scalar is emitted from its own effective vertex, at the same mass m. We computed them with the code of Ref. [7], replacing the CP-odd vertices of O_G by those of the CP-even operator $H\, G^a_{\mu\nu} G^{a\mu\nu}$ and leaving the rest of the calculation unchanged. The code reproduces the one-loop AA result of that reference to 40 digits, and the HH double poles agree with Catani's formula [32].

In the conventions used here, the HH tree amplitude at threshold is $-2im^2$, opposite in sign to the AA amplitude. The corresponding forward coefficient satisfies

$$D(\beta) = -F_S\left(\frac{1+\beta}{2}\right) \qquad (gg \to HH)$$

to a relative accuracy of 3×10^{-15} for $0.001 \le \beta \le 0.7$. The normalisation fixed from AA therefore also reproduces the HH logarithm without an additional fitted constant. In units of the respective tree amplitudes, the ratio of the scalar and pseudoscalar logarithmic coefficients at threshold is

$$\frac{F_S(\frac{1}{2})}{F_A(\frac{1}{2})} = \frac{9}{7}. \tag{47}$$

The opposite helicity amplitude again has no logarithm. Its extracted forward coefficient is about 3×10^{-14} times that of $\mathcal{M}_1$. For this check, the diagram containing the four-gluon term of the CP-even operator and the contact counterterm for two such insertions were not included. Neither depends on p_T, so neither changes the logarithmic coefficient being tested.

Figure 4(a) compares the extracted coefficients with the two helicity weights. The AA points lie on the curve F_A for both $(N_c, n_f) = (3,5)$ and $(4,2)$, so that after the normalisation of Eq. (46) the coefficient does not depend on N_c or n_f. The HH points (triangles), plotted with their sign reversed, lie on the curve F_S, which is separated from F_A by

$$F_S - F_A = 4x(1-x) = 1 - \beta^2. \tag{48}$$

Panel (b) shows the relative deviations from the predictions; the two-loop points are discussed in Sec. IV D.

e. Colour, scale dependence and phase. The factorisation also explains the flavour and colour dependence of the one-loop logarithm. The O_G vertices couple only to gluons, so only gluon splittings enter this singular configuration and its coefficient is independent of n_f. The factor N_c comes from the colour contraction of the two three-gluon vertices. Renormalisation and IR subtraction do not change this logarithmic coefficient: at one loop, their contributions are proportional to the tree amplitude and do not generate a logarithm of p_T. The coefficient is therefore independent of μ_R.

Finally, the coefficient $7\pi N_c\, m_A^2$ of $\mathcal{L}$ is real, whereas $\mathcal{M}_1^{(0)} \to 2im_A^2$ at threshold. The logarithm is thus out of phase with the tree amplitude by 90°. This relative phase is consistent with the absorptive origin of the one-loop DPS singularity, which can be obtained from cuts that put internal lines on shell [2]. It explains why the one-loop DPS logarithm does not contribute to the interference with the tree amplitude, as discussed in Sec. V.

D. Two loops

The leading logarithms of the calculated two-loop coefficient $\mathcal{M}_1$ can be written in terms of the one-loop DPS logarithm. Comparing the coefficients of Eq. (40) with the coefficient $7\pi N_c$ of Eq. (42) gives, for general N_c and n_f,

$$\widehat{A}_{13}\mathcal{L}^3 + \widehat{A}_{12}\mathcal{L}^2 = 7\pi N_c\, \mathcal{L}\left[-\tfrac{4N_c}{3}\,\mathcal{L}^2 - 2\beta_0\,\mathcal{L}\right], \tag{49}$$

with β_0 as in Eq. (24). A similar relation holds in planar $\mathcal{N} = 4$ super Yang–Mills theory. Reference [3] studied the six-gluon maximally helicity violating (MHV) amplitude in planar $\mathcal{N} = 4$ super Yang–Mills theory, which has two negative and four positive helicities when all momenta are outgoing. Near the configuration that mimics double parton scattering, its singular terms obey an evolution equation, derived following the analysis of crossing Wilson lines by Korchemskaya and Korchemsky [47],

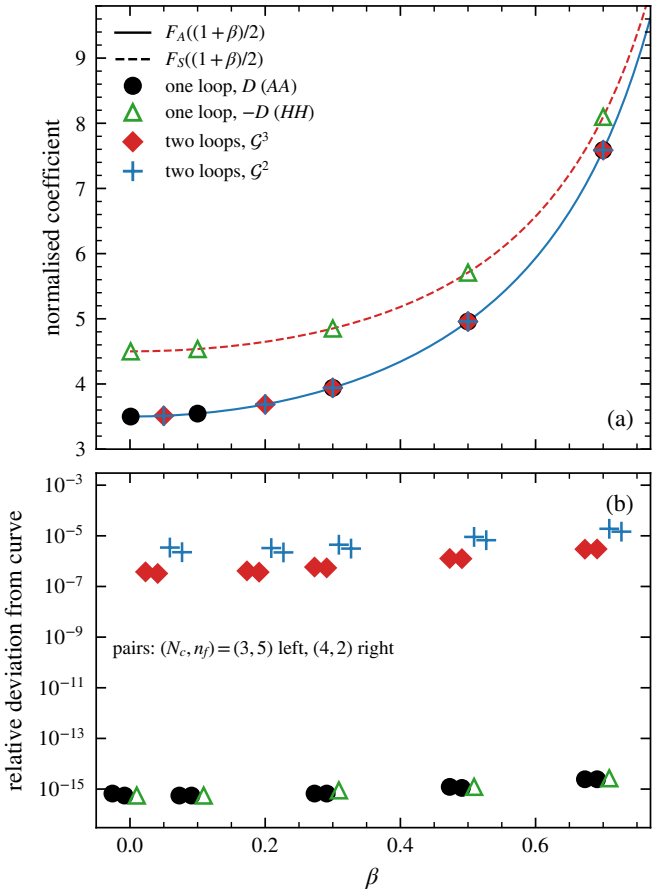


FIG. 4. Forward coefficients of the DPS logarithms and the helicity weights; each coefficient has the same symbol in both panels. (a) $(N_c, n_f) = (3,5)$. Circles: $D(\beta)$ for $gg \to AA$, Eq. (46). Triangles: $-D(\beta)$ for $gg \to HH$ through two CP-even effective vertices; the sign reversal removes the relative sign of the scalar and pseudoscalar tree amplitudes at threshold. Diamonds and crosses: the real parts of the fitted coefficients of $\mathcal{G}^3$ and $\mathcal{G}^2$ in the forward limit of the two-loop remainder of $gg \to AA$, fitted over $1-c = 10^{-7}, \ldots, 10^{-10}$ and divided by $-\frac{4}{3}N_c \cdot 2\pi N_c m^2$ and $-2\beta_0 \cdot 2\pi N_c m^2$ (Sec. IV D). Solid and dashed curves: $F_A((1+\beta)/2)$ and $F_S((1+\beta)/2)$. The values for $(N_c, n_f) = (4,2)$ coincide with those shown. (b) Relative deviation of each point from its curve in (a), for $(N_c, n_f) = (3,5)$ and, for $gg \to AA$, $(4,2)$; the symbols are displaced horizontally for legibility.

which fixes the leading logarithms to all orders. The ratio of the leading two-loop and one-loop terms, including their powers of the coupling, is $-(a/6)\ln^2\delta$, where $a = g^2 N_c/(8\pi^2)$ and δ, which measures the approach to the singular configuration, is proportional to the squared transverse momentum at fixed values of the other kinematic variables. With $a = 2N_c a_s$ and $\ln\delta = 2\mathcal{L} + \text{const}$, the ratio becomes $-\frac{4}{3}N_c a_s \mathcal{L}^2$ at leading logarithmic order. This is the ratio of the cubic term in Eq. (49) to the one-loop logarithm, once the additional power of a_s at two loops is included. The second term, $-2\beta_0\mathcal{L}$, has no counterpart in the $\mathcal{N} = 4$ theory; the number of quark flavours enters it only through β_0.

The same β_0 term appears in the NLO momentum-space kernel for the splitting $g \to gg$ in unpolarised colour singlet double parton distributions [23], in which the logarithms are of the transverse momentum mismatch Δ between amplitude and conjugate amplitude, not of the transverse momentum of a produced particle; the comparison concerns the coefficients of the logarithms only. Equations (50) and (135) of Ref. [23] give the coefficient of $\ln^2(\mu^2/\Delta^2)$ in units of $\alpha_s/2\pi$; relative to the LO term, its β_0 part corresponds to $-2\beta_0\mathcal{L}$ in the normalisation of a_s used here. Half of it comes from the renormalisation of the coupling and half from the δ-function parts of the DGLAP kernels, so that the running of the coupling alone would give $-\beta_0\mathcal{L}$. The other terms of the kernel are not part of this comparison: a term proportional to $C_A \ln[x(1-x)]$ from the plus distributions, which contributes where the two momentum fractions add up to one, and the real emission terms, which contribute where they add up to less than one.

The following tests indicate that the relation of Eq. (49) extends beyond threshold. In the forward limit at fixed velocity a convenient variable is $\mathcal{G} = \frac{1}{2}\ln[(m_A^4 - tu)/m_A^4] = \mathcal{L} - \ln(1-\beta_A^2)$, in which the one-loop DPS logarithm is $C_1\mathcal{G}$ and the relation reads

$$\mathcal{M}_1^{(2)} \supset C_1\Big[-\frac{4}{3}N_c\,\mathcal{G}^3 - 2\beta_0\,\mathcal{G}^2\Big], \qquad (50)$$
$$C_1 = 2\pi N_c m_A^2 F_A\Big(\frac{1+\beta_A}{2}\Big),$$

with C_1 from Eq. (46). In the threshold series, Eq. (50) holds exactly through β_A^6 for general N_c and n_f; at fixed angle, the coefficient of $\mathcal{L}^3$ follows the expansion of $F_A\big((1+\beta_A)/2\big)$ order by order.

Away from threshold we evaluated the released two-loop remainder of $\mathcal{M}_1$ at $1-c = 10^{-6}, \ldots, 10^{-10}$ with 180-digit arithmetic, checked with 220 digits, which agree to at least 82 digits, and fitted a cubic polynomial in $\ln(1-c)$. For $0.05 \le \beta \le 0.7$ and $(N_c, n_f) = (3,5)$ and $(4,2)$, the real parts of the fitted coefficients of $\mathcal{G}^3$ and $\mathcal{G}^2$ agree with Eq. (50) to 4×10^{-6} and 2×10^{-5} (Fig. 4), and their imaginary parts are below 2×10^{-5} of their magnitudes, when the point at $1-c = 10^{-6}$ is left out; with it, the deviations are about four times larger, as expected from power corrections in $1-c$. Within the range tested, the two-loop leading logarithms of $\mathcal{M}_1$ in the forward limit are therefore the one-loop DPS logarithm, with its helicity weight $F_A(x)$, multiplied by the same factor at every velocity.

The $|J_z| = 2$ amplitude acquires a logarithm at two loops: the term $\widehat{A}_{22}\mathcal{L}^2$, with $\widehat{A}_{22} = (-4N_c)\cdot\frac{1}{2}\pi N_c$, is the one-loop constant multiplied by $-4N_c\,\mathcal{L}^2$. The forward-limit calculation described next indicates that the same coefficient relation persists away from threshold. In the forward limit $c \to 1$ at fixed velocity, where $p_T \to 0$ as well, we evaluated the released two-loop remainder of $\mathcal{M}_2$ at $1-c = 10^{-5}$, 10^{-6} and 10^{-7} with 140-digit arithmetic and fitted a quadratic polynomial in $\ln(1-c)$; a repetition

with 180 digits agrees to at least 51 digits. For $0.05 \le \beta \le 0.5$ and $(N_c, n_f) = (3,5)$ and $(4,2)$, the real part of the coefficient of $\ln^2(2p_T/m)$ equals $-4N_c$ times the forward limit of $\mathcal{M}_2^{(1)}$ to 6×10^{-5}, although that limit varies by a factor of 3.4 over this range. Its imaginary part, which vanishes in the proposed coefficient relation, is below 2×10^{-4} of its magnitude, and including the point at $1-c=10^{-4}$ in the fit increases the deviation to 7×10^{-4}, as expected from power corrections. Within the range tested, the one-loop amplitude and the fitted two-loop logarithm are therefore consistent with

$$a_s\mathcal{M}_2^{(1)}\left[1-4N_c a_s\ln^2(2p_T/m)\right], \tag{51}$$

with no additional velocity or flavour dependence resolved in the coefficient of the double logarithm.

The observed combination has the form of the expansion of $\exp(-4N_c a_s\mathcal{L}^2)$ to first order in a_s. Multiplied by the one-loop coefficient $7\pi N_c$ and integrated over $\mathcal{L}$, the same exponential gives the cubic term of Eq. (49). At lowest order it also corresponds to the factor $\exp(-\gamma_K\ln^2\delta/8)$ of Ref. [3], with the cusp anomalous dimension $\gamma_K = 4a$. However, two loops cannot decide whether this factor exponentiates in QCD. The J_z argument of Sec. IV C concerns the one-loop singularity and does not forbid this logarithm. Nor can it be explained by loop corrections to the $gg\to A$ subprocess amplitudes, since conservation of angular momentum forbids $gg\to A$ for opposite gluon helicities to all orders; its dynamical origin remains open. A related effect occurs in planar $\mathcal{N}=4$ super Yang–Mills theory. The next-to-MHV (NMHV) six-gluon amplitude, with three negative helicities, has no such singularity at one loop because of J_z conservation, but Dixon and Esterlis found logarithmic singularities starting at two loops, in the region where the limit overlaps with multi-Regge kinematics, and suggested that they may come from contributions in which more than one particle crosses a cut [3].

The forward limit of $\mathcal{M}_2^{(1)}$ itself has a closed form. The exact one-loop remainders agree with

$$\lim_{c\to1}\frac{\mathcal{M}_2^{(1)}}{m^2} = \pi N_c\,\frac{x^4+(1-x)^4}{x(1-x)}, \qquad x=\frac{1+\beta}{2}, \tag{52}$$

to 2×10^{-34} for $0.05\le\beta\le0.9$ and both sets of (N_c,n_f); at threshold it gives $\frac{1}{2}\pi N_c$. With the weights of Eq. (43), the fraction in Eq. (52) equals $w_{+-}(x)w_{-+}(1-x)+w_{-+}(x)w_{+-}(1-x)$: the helicity sum of Sec. IV C for the two pairings that an opposite-helicity initial state allows when each effective vertex receives two gluons of equal helicity, with the same overall factor πN_c as in Eq. (46). The one-loop constant of $\mathcal{M}_2$ thus has the form expected if it arose from the DPS region, where the azimuthal phase of these pairings removes the logarithm and leaves a finite term; we have not derived this from the factorisation formula. The two helicity amplitudes therefore provide an explicit two-loop QCD benchmark for a more general description of this singular kinematic region.

Figure 3 compares the three orders. With $a_s = 0.0094$ ($\alpha_s = 0.118$), at $\beta_A = 0.1$ and $c = 1/3$, the one- and two-loop terms of $\mathcal{M}_1$ are 1.1 and 0.7 times the tree term, and the one-loop term of $\mathcal{M}_2$ is 2.3 times the tree term. The last ratio grows as β_A^{-2} towards threshold, since the tree-level $|J_z|=2$ amplitude vanishes there. The one-loop terms include the scale term of Eq. (24). Towards threshold, the loop terms of $\mathcal{M}_1$ grow with $|\mathcal{L}|$, the two-loop term as $|\mathcal{L}|^3$ and the one-loop term as $|\mathcal{L}|$, so that below $\beta_A\approx1.7\times10^{-3}$ the two-loop term exceeds the one-loop term at this angle. Because the logarithms are out of phase with the tree amplitude, the strongest of them are absent from the finite virtual correction (Sec. V). In panel (b) the one-loop term of $\mathcal{M}_2$ tends to the constant of Eq. (42), while the two-loop term grows as $\mathcal{L}^2$, the logarithm that the $|J_z|=2$ amplitude acquires at two loops.

V. CONSEQUENCES FOR THE CROSS SECTION

A. Leading order

Averaging Eq. (18) over helicities and colours, and including a factor $\frac{1}{2}$ for identical particles, gives the LO partonic cross section of the Type II amplitude,

$$\frac{\mathrm{d}\hat\sigma}{\mathrm{d}c} = \frac{\beta_A\,C_G^4}{128\pi\hat s\,(N_c^2-1)}\left(|\mathcal{M}_1^{(0)}|^2+|\mathcal{M}_2^{(0)}|^2\right) \tag{53}$$

for $-1<c<1$. For $\alpha_s=0.118$, $\cot\beta=1$ and $m_A=125\,\mathrm{GeV}$, we find $\hat\sigma = 1.29\times10^{-8}\,\mathrm{pb}$ at $\beta_A=0.1$ (Fig. 5(a)). The cross section rises as β_A above threshold and as $\hat s$ at large $\hat s$, as expected for dimension-five operators, and it is of fourth order in a_s through C_G^4. In a 2HDM, it is a small part of the total, which is dominated by Type I (Sec. II E).

B. Finite virtual corrections

To examine the effect of the loop corrections on the Type II squared amplitude, we define two dimensionless coefficients, V_1 and V_2, relative to the tree contribution. Let $\mathcal{A}_{\mathrm{fin}}$ denote the amplitude constructed from the finite remainders of Eq. (22), in the normalisation of Eq. (24). Using the helicity decomposition in Eq. (18), its square

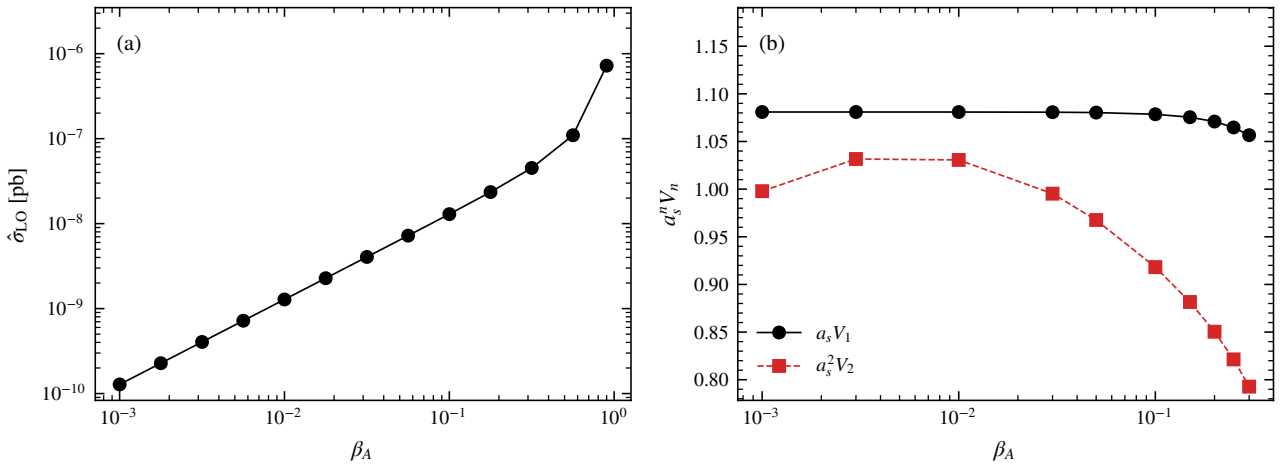


FIG. 5. (a) Leading order partonic cross section of the Type II amplitude, Eq. (53), for $m_A = 125\,\text{GeV}$, $\alpha_s = 0.118$ and $\cot\beta = 1$. (b) Angle averages $a_s\langle V_1\rangle$ and $a_s^2\langle V_2\rangle$ of the finite virtual corrections, Eq. (55), for $\mu_R = m_A$, $N_c = 3$ and $n_f = 5$.

can be written as

$$\sum_{\text{hel,col}} |\mathcal{A}_{\text{fin}}|^2 = 2(N_c^2-1)\,C_G^4 \sum_i |\mathcal{M}_i^{(0)}|^2 \times \left[1 + a_s V_1 + a_s^2 V_2 + \mathcal{O}(a_s^3)\right], \text{ where}$$
$$V_1 = \frac{2\,\text{Re}\sum_i \mathcal{M}_i^{(0)*}\mathcal{M}_i^{(1)}}{\sum_i |\mathcal{M}_i^{(0)}|^2},$$
$$V_2 = \frac{\sum_i \left(2\,\text{Re}\left[\mathcal{M}_i^{(0)*}\mathcal{M}_i^{(2)}\right] + |\mathcal{M}_i^{(1)}|^2\right)}{\sum_i |\mathcal{M}_i^{(0)}|^2}. \tag{54}$$

The sums run over the two tensor coefficients, $i = 1, 2$. The coefficient V_1 contains the interference between the tree and one-loop amplitudes. At the next order, V_2 contains both the interference between the tree and two-loop amplitudes and the square of the one-loop amplitude. These coefficients refer only to the O_G–O_G Type II contribution; they are the NLO and NNLO finite virtual terms when the Type II tree amplitude is the leading amplitude (Sec. II E). The real emission contributions are not included here. The quantities V_1 and V_2 therefore retain the dependence on the subtraction scheme of Eq. (22), which cancels only when the real emission and the counterterms for initial state collinear factorisation are added; they are not observables or K factors. The coefficient of the squared logarithm derived below is not affected by the finite changes of the subtraction scheme specified below.

At fixed β_A, both coefficients depend on the scattering angle. To display their velocity dependence in Fig. 5(b), we average them over the angle with the Type II LO cross section as the weight. Since the remaining factors in Eq. (53) depend only on β_A and cancel in this average, $a_s\langle V_1\rangle$ and $a_s^2\langle V_2\rangle$ are the finite virtual contributions relative to the angle-integrated tree contribution, rather than unweighted averages. The weighted average is

$$\langle V_n\rangle = \frac{\displaystyle\int_{-1}^{1} \mathrm{d}c \sum_i |\mathcal{M}_i^{(0)}|^2\, V_n}{\displaystyle\int_{-1}^{1} \mathrm{d}c \sum_i |\mathcal{M}_i^{(0)}|^2}, \qquad n = 1, 2. \tag{55}$$

The angular integrals are evaluated with 64-point Gauss–Legendre quadrature; with 128 points, $\langle V_1\rangle$ and $\langle V_2\rangle$ change by relative amounts smaller than 4×10^{-52} and 10^{-5} at $\beta_A = 0.01$, 0.1 and 0.3. We evaluate the one-loop remainders without a threshold expansion and the two-loop remainders with the series through β_A^6. For $\alpha_s = 0.118$ and $\mu_R = m_A = 125\,\text{GeV}$, the curves of Fig. 5(b) lie between 1.06 and 1.08 for $a_s\langle V_1\rangle$, and between 0.79 and 1.03 for $a_s^2\langle V_2\rangle$, over $10^{-3} \le \beta_A \le 0.3$. At fixed kinematics, the scale dependence of the one-loop remainder in Eq. (24) gives $\partial V_1/\partial \ln\mu_R^2 = 4\beta_0$.

a. One-loop contribution. The threshold behaviour of V_1 and V_2 follows from the relative phases of the tree and loop amplitudes. At fixed scattering angle, $\mathcal{M}_1^{(0)} \to 2im_A^2$, whereas $\mathcal{M}_2^{(0)}$ vanishes as β_A^2. The squared tree amplitude in the denominator of Eq. (54) therefore tends to $4m_A^4$, and at leading power interference with the tree amplitude selects the imaginary part of $\mathcal{M}_1^{(1)}$ or $\mathcal{M}_1^{(2)}$. With $\ell = \ln(2p_T/m_A)$, Eq. (36) gives $\mathcal{L} = \ell + i\pi/2 + \mathcal{O}(\beta_A^2)$, and the one-loop result of Eq. (42) reads

$$\frac{\mathcal{M}_1^{(1)}}{m_A^2} = 7\pi N_c\,\ell + \frac{i\pi}{2}7\pi N_c + \widehat{A}_{10}^{(1)}.$$

The coefficient of ℓ is real, so the logarithm drops out of the interference with the imaginary tree amplitude. Only the imaginary constants remain:

$$V_1 \longrightarrow \frac{\pi}{2}7\pi N_c + \text{Im}\,\widehat{A}_{10}^{(1)}. \tag{56}$$

For $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$, this gives $V_1 \to 115.11$. The one-loop remainder thus contains the DPS logarithm, but the leading power finite virtual correction does not. The limit is also independent of the angle, which is why the angular average in Fig. 5(b) is nearly constant.

b. Two-loop contribution. The interference of the two-loop amplitude with the tree amplitude again selects the imaginary part of the loop remainder, but the square of the one-loop amplitude retains the real logarithm. Using Eqs. (39) and (42), the leading power terms of Eq. (54) are

$$V_2\big|_{\beta_A^0} = \mathrm{Im}\left[\sum_{k=0}^{3} \widehat{A}_{1k}\mathcal{L}^k\right] + \frac{1}{4}\left|7\pi N_c\,\mathcal{L} + \widehat{A}_{10}^{(1)}\right|^2 + \frac{\pi^2 N_c^2}{16}. \tag{57}$$

The three terms are the interference of the two-loop amplitude with the tree amplitude, the square of $\mathcal{M}_1^{(1)}$, and the square of $\mathcal{M}_2^{(1)} \to \pi N_c m_A^2/2$. The last contributes at leading power although the corresponding tree amplitude vanishes, while the interference of $\mathcal{M}_2^{(0)}$ with $\mathcal{M}_2^{(2)}$ is suppressed by β_A^2, up to logarithms.

The coefficients $\widehat{A}_{13}$ and $\widehat{A}_{12}$ are real, and at leading power

$$\mathrm{Im}\,\mathcal{L}^3 = \frac{3\pi}{2}\ell^2 - \frac{\pi^3}{8}, \qquad \mathrm{Im}\,\mathcal{L}^2 = \pi\ell.$$

The cubic logarithm of the two-loop amplitude therefore gives only a quadratic logarithm in its interference with the tree amplitude, and $\widehat{A}_{12}\mathcal{L}^2$ only a linear one. The square of the one-loop amplitude is at most quadratic in ℓ. The leading threshold expansion of V_2 therefore has the form

$$V_2 = v_0 + v_1\ell + v_2\ell^2 + \mathcal{O}_{\log}(\beta_A^2), \tag{58}$$

where the subscripts of v_0, v_1 and v_2 give the power of ℓ, not the number of loops.

The coefficient v_2 receives two contributions: $\frac{3}{2}\pi\widehat{A}_{13} = -14\pi^2 N_c^2$ from the interference of the two-loop amplitude with the tree amplitude, and $\frac{1}{4}(7\pi N_c)^2 = \frac{49}{4}\pi^2 N_c^2$ from the square of the one-loop amplitude. Their sum is

$$v_2 = -14\pi^2 N_c^2 + \frac{49}{4}\pi^2 N_c^2 = -\frac{7}{4}\pi^2 N_c^2. \tag{59}$$

The two contributions partly cancel. At leading power, the DPS logarithm therefore first appears in the finite virtual correction in V_2, with highest power ℓ^2.

A fit of the numerical values of V_2 at $c = 1/3$ over $10^{-8} \le \beta_A \le 10^{-6}$ gives $v_2 = -155.446269$, while Eq. (59) gives -155.446269 for $N_c = 3$; the relative difference before rounding is 3×10^{-11}. For $a_s = 0.0094$, the quadratic term in the finite virtual correction is $a_s^2 v_2 \ell^2 = -0.0137\,\ell^2$.

c. Remaining coefficients and numerical behaviour. At leading power, v_0, v_1 and v_2 are independent of the scattering angle. The angular dependence of V_2 itself remains through ℓ. Fits for v_0 and v_1 at $c = 0$, $\frac{1}{3}$ and $\frac{2}{3}$ agree within a relative spread of 5×10^{-11}. The coefficient of the linear logarithm follows from the same expansion:

$$\begin{aligned} v_1 &= \pi\widehat{A}_{12} + \mathrm{Im}\,\widehat{A}_{11} + \frac{7\pi N_c}{2}\,\mathrm{Re}\,\widehat{A}_{10}^{(1)} \\ &= \frac{7}{2}\pi N_c\,\mathrm{Re}\,\widehat{A}_{10}^{(1)} + 7\pi^2 N_c\beta_0. \end{aligned}$$

For the benchmark parameters, this gives $v_1 = -1508.5$, in agreement with the numerical fit to a relative difference of 1×10^{-10}. The coefficient is independent of μ_R, since $\mathrm{Re}\,\widehat{A}_{10}^{(1)}$ is scale independent. The same is true of v_2. At leading power, the explicit scale dependence of V_2 is therefore confined to v_0, which changes from 8118 to 15421 when μ_R is increased from m_A to $2m_A$.

Although ℓ^2 is the highest power of the logarithm, it need not give the largest contribution at the velocities considered. At $c = 1/3$, the ratio $(v_1\ell + v_2\ell^2)/V_2$ is approximately 20% at $\beta_A = 0.1$ and 30% at $\beta_A = 0.01$. The quadratic term alone contributes -4% and -21% of the total V_2, respectively. Since $\ell < 0$ in this region and both v_1 and v_2 are negative, the linear term is positive and the quadratic term is negative. This competition also explains the shape of the correction as a function of velocity. The leading power expression has a maximum at

$$\ell_{\max} = -\frac{v_1}{2v_2} = -4.85.$$

At $c = 1/3$, this corresponds to $\beta_A \simeq 0.0041$. For larger β_A, ℓ increases and the leading power value of V_2 decreases. The same balance between the linear and quadratic terms explains the turnover of the angular average $a_s^2\langle V_2\rangle$ in Fig. 5(b). At $c = 1/3$ and $\beta_A = 0.1$, including the terms beyond leading power changes V_2 by only 0.3%.

d. DPS interpretation and subtraction dependence. The two contributions to v_2 also differ in their relation to the DPS configuration. In the square of the one-loop amplitude, the DPS region occurs in both the amplitude and its complex conjugate, that is, on both sides of the final state cut. This is the configuration associated with perturbative splittings in both protons that overlaps with DPS and requires a subtraction to avoid double counting [25, 29]. In the interference with the tree amplitude, the DPS region occurs only on the loop side of the cut. Both contributions enter the finite virtual coefficient V_2 defined in Eq. (54); the logarithmic term in its one-loop square alone would therefore not give the complete coefficient v_2.

Although V_2 depends on the IR subtraction convention, v_2 is unchanged by finite scheme changes that multiply lower order amplitudes by angle independent functions regular at the pair threshold. The tree amplitude has no logarithm, and the one-loop remainder contains

at most one power of ℓ. Such changes can therefore affect the constant and linear terms in V_2, but not the coefficient of ℓ^2.

Finally, with the Wilson coefficient counting specified in Sec. II E, the one-loop O_G–O_J amplitude contributes at the same order as the two-loop O_G–O_G amplitude. Its coefficient C_J depends on m_t in the heavy top quark theory [7]. By Eq. (25), the mixed amplitude is proportional to the tree amplitude. After including its Wilson coefficient and matching the normalisation, its interference with the tree amplitude therefore gives a kinematically constant shift to V_2. It changes v_0, but not v_1 or v_2, and is not included in the results above.

C. Use in higher order calculations

For the two-body final state, $m_{AA} = \sqrt{\hat{s}} = 2m_A/\sqrt{1-\beta_A^2}$. The range $\beta_A \leq 0.2$ therefore covers pair invariant masses up to 2.06% above threshold; for $m_A = 125\,\text{GeV}$, this is $m_{AA} \leq 255.2\,\text{GeV}$. The point $\beta_A = 0.3$ lies 4.83% above threshold. I recommend the series through β_A^6 for $0 < \beta_A \leq 0.2$ at the angles tested in Tables II and IV; at $\beta_A = 0.3$, the largest relative error of the complete coefficients on the validation grid is 1.94×10^{-3}. These checks do not establish a uniform accuracy bound as $c \to \pm 1$, where the logarithms diverge. The series retain the complex values of both tensor coefficients, so they can be used in interference with other amplitudes, provided these are combined with the same normalisation, Wilson coefficient conventions, tensor basis and infrared subtraction scheme.

One application is the hard function of a threshold resummed or soft-virtual prediction for the pair invariant mass distribution. There, threshold refers to the limit of little energy available for additional radiation at fixed m_{AA}, not to $\beta_A \to 0$ [17, 18, 20]. At leading power in the soft limit, the soft and collinear factors are universal for gluon fusion into a colour singlet, and the process dependence enters through the finite virtual contributions at each m_{AA} [19, 21]; the series supply the two-loop Type II input near $m_{AA} = 2m_A$. When the Type II tree amplitude is the leading amplitude, the required combinations are those of Eq. (54), and integrated over the angle at fixed m_{AA} they give the averages of Eq. (55); Ref. [20] gives the corresponding construction for Higgs pairs.

In a 2HDM, where these remainders first contribute at N^{3}LO through interference with the leading Type I amplitude (Sec. II E), the averages of Eq. (55) cannot be used directly, since the interference with Type I must be formed before the angular integration, and the remaining amplitudes, including any mixed O_G–O_J contribution, must be included at the same order. For Higgs pairs, similarly, the diagrams with two effective vertices enter the NNLO virtual corrections only at tree level and at one loop [48], and their two-loop terms are an ingredient of N^{3}LO [9].

VI. CONCLUSIONS

We have expanded the two-loop Type II finite remainders for $gg \to AA$ with two O_G insertions about the pair production threshold through β_A^6 at fixed scattering angle. After the endpoint constants are expressed in a basis of eight transcendental constants, all inverse powers of β_A and the odd powers through β_A^5 cancel identically, and every coefficient is even in $\cos\theta$. For $0 < \beta_A \leq 0.2$ the series reproduce the two complete coefficients to 3.23×10^{-6} and 6.77×10^{-5} at the points tested, including the forward angles, in files a factor of 5061 smaller than those of Ref. [7]. The remainders of that reference are normalised differently at different orders, with a factor 64 at two loops and $\mu_R = 1$ GeV at one loop; after conversion to a common normalisation they satisfy the renormalisation group relations of Eqs. (B1) and (B3) (App. B).

At leading power the angular dependence enters only through the logarithm of the transverse momentum, which we identified with the DPS singularity of Ref. [2]. The $g \to gg$ splitting weights and the relative sign of the CP-odd subprocess amplitudes fix the momentum fraction dependence of its coefficient; with the overall constant fixed at threshold, this reproduces the exact one-loop coefficient in the forward limit to 3×10^{-15}, and a one-loop $gg \to HH$ calculation confirms the CP-even weight with the same constant. At two loops the $J_z = 0$ amplitude grows as $\ln^3 p_T$, with the leading logarithmic ratio of planar $\mathcal{N} = 4$ super Yang–Mills theory [3] and a β_0 term with the structure of the NLO $g \to gg$ splitting kernel [23]; in the forward limit both logarithms follow the one-loop helicity weight at all velocities tested. The $|J_z| = 2$ amplitude, which has no one-loop logarithm, grows as $\ln^2 p_T$, and the forward fits identify the real part of its coefficient, within the velocities tested, as $-4N_c$ times the one-loop forward amplitude of Eq. (52). In the finite virtual corrections the one-loop logarithm drops out because it is out of phase with the tree amplitude, and at two loops the cubic logarithm cancels, leaving $-\frac{7}{4}\pi^2 N_c^2 \ln^2(2p_T/m_A)$ in V_2, which results from the combination of the interference of the two-loop amplitude with the tree amplitude and the square of the one-loop amplitude.

Several questions remain open: the derivation of the overall factor $\pi N_c m^2$ from the factorisation formula with the effective vertices, the mechanism that produces the $|J_z| = 2$ double logarithm, and whether the observed factor exponentiates. A general treatment of this singular region would have to reproduce the different logarithmic powers of the two amplitudes, their colour and flavour dependence and the reduced singularity of the finite virtual correction; the present results provide explicit two-loop QCD benchmarks for it, and the known two-loop $gg \to HH$ amplitudes with two effective vertices [9] permit the same comparison for the scalar case. If the one-loop factorisation applies to the full theory, the on-shell triangle form factors of Eq. (27) multiply the logarithmic coefficient by 1.096 at $m_A = 125$ GeV; this does not

determine the mass dependence of the other terms.

The series provide the two-loop Type II input to the hard function for the pair invariant mass distribution near $m_{AA} = 2m_A$, at NNLO for a pseudoscalar coupled to gluons only through $G\tilde{G}$ and at N^{3}LO in a 2HDM (Sec. V C). The series, the evaluator, the validation records and the code of the construction are provided with the paper [30].

DATA AVAILABILITY

The threshold series, construction files, validation records, the code of the construction and validation, the evaluator, the one-loop $gg \to HH$ coefficients of Sec. IV C and the scripts that reproduce all tables and figures form a self-contained package. Version 7.3.3 is provided as ancillary files with the arXiv version of this article and as Supplemental Material [30]; the package is maintained at `https://github.com/abhattacharya1992/ggAA-threshold`. The unexpanded finite remainders used as input were released as ancillary files with Ref. [7]; the package downloads them from arXiv and verifies their checksums. The code used for the one-loop $gg \to HH$ amplitudes of Sec. IV C is not included; it is available from the author on request.

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Appendix A: Endpoint constants

The boundary values of the expansion contain the constants

$$\mathrm{MB}[p,q,a,w] = \frac{a}{(p-1)!}\int_0^1 \mathrm{d}t\,\frac{[-\ln t]^{p-1}}{1-at}\,\mathrm{Li}_q(wt), \tag{A1}$$

with $\mathrm{Li}_1(z) = -\ln(1-z)$. Here p and q are positive integers, $p-1$ being the power of the logarithm and q the weight of the polylogarithm, so that $\mathrm{MB}[p,q,a,w]$ has weight $p+q$; the constants that occur have $p,q \in \{1,2\}$, and a and w take the values -1, $\frac{1}{2}$ and 1. Besides the elementary values $\mathrm{MB}[2,2,1,1] = \pi^4/120$ and $\mathrm{MB}[2,1,1,1] = \zeta_3$, thirteen distinct constants occur. They were evaluated by quadrature at high precision, independently of the amplitude data, and reduced to the basis of Eq. (30) with the PSLQ algorithm [43] at 110 digits. Table VII lists the results; the largest residual is 4×10^{-107}, and the same relations were obtained in Mathematica.

TABLE VII. Reductions of the endpoint constants of Eq. (A1), found with PSLQ at 110 digits; the largest residual is 4×10^{-107}.

$[p,q,a,w]$	reduction
$[1,2,-1,1]$	$-\frac{\pi^2\ln 2}{6}+\frac{5\zeta_3}{8}$
$[1,2,1/2,1/2]$	$-\frac{\zeta_3}{4}-\frac{\ln^3 2}{6}+\frac{\pi^2\ln 2}{12}$
$[1,2,1/2,1]$	$-\zeta_3+\frac{\pi^2\ln 2}{4}$
$[2,1,-1,-1]$	$\frac{\zeta_3}{8}$
$[2,1,-1,1]$	$-\frac{13\zeta_3}{8}+\frac{\pi^2\ln 2}{4}$
$[2,1,1,-1]$	$-\frac{\pi^2\ln 2}{4}+\zeta_3$
$[2,1,1,1/2]$	$-\frac{5\zeta_3}{8}+\frac{\pi^2\ln 2}{6}$
$[2,1,1/2,1/2]$	$-\frac{\ln^3 2}{6}+\frac{\zeta_3}{8}$
$[2,1,1/2,1]$	$-\frac{\pi^2\ln 2}{4}+\frac{13\zeta_3}{8}$
$[2,2,1,-1]$	$-\frac{71\pi^4}{1440}-\frac{\pi^2\ln^2 2}{6}+\frac{\ln^4 2}{6}+4\,\mathrm{Li}_4(\frac{1}{2})+\frac{7\ln 2\zeta_3}{2}$
$[2,2,1,1/2]$	$-\mathrm{Li}_4(\frac{1}{2})-\frac{\ln 2\zeta_3}{4}-\frac{\pi^2\ln^2 2}{24}-\frac{\ln^4 2}{24}+\frac{19\pi^4}{1440}$
$[2,2,1/2,1/2]$	$-\frac{\pi^2\ln^2 2}{24}+\frac{\ln^4 2}{24}+\frac{\pi^4}{1440}+\frac{\ln 2\zeta_3}{4}$
$[2,2,1/2,1]$	$-3\,\mathrm{Li}_4(\frac{1}{2})-\frac{\pi^2\ln^2 2}{8}-\frac{\ln^4 2}{8}+\frac{7\pi^4}{288}$

Appendix B: Normalisation of the released remainders

Six checks establish the normalisation of Eq. (24) for the tree, one-loop and two-loop amplitudes. They use the tree amplitudes and the one- and two-loop remainders of Ref. [7], together with the Mathematica code of the original calculation.

Tree level. The projections of the two tree diagrams from the original calculation give Eq. (20), after removing a colour factor N_c^2-1 that is included in the file for T_1 but not in the file for T_2. A direct evaluation of the two diagrams with the vertex $iC_G\varepsilon_{\mu\alpha\rho\beta}k_1^\mu k_2^\rho\delta^{ab}$, which follows from Eq. (1), gives $|\mathcal{A}_{\pm\pm}| = C_G^2|\mathcal{M}_1^{(0)}|$ and $|\mathcal{A}_{\pm\mp}| = C_G^2|\mathcal{M}_2^{(0)}|$ to 14 digits.

Tree and one loop. The coefficient of n_f in $\mathcal{M}_i^{(1)}$, divided by $\mathcal{M}_i^{(0)}$, is the same for $i=1,2$ at threshold, and its imaginary part equals $-2\pi/3$ to 15 digits at every tested point. This is the value that the subtraction term $2I^{(1)}\mathcal{M}^{(0)}$ of Eq. (22) produces through the n_f part of β_0 multiplying $\ln(-\hat{s}/\mu_R^2)$; the one-loop diagrams proportional to n_f have no cut in the $\hat{s}$ channel. This check fixes the normalisation and phase of the tree amplitude relative to the one-loop remainders.

Scale dependence at one loop. Apart from the overall factor m_A^2, a remainder with the dimension of m_A^2 can depend on m_A and μ_R only through their ratio. The released one-loop remainders do not have this property: when m_A and μ_R are multiplied by a common factor λ at fixed β_A and c, $\mathcal{M}^{(1)}_{i,\mathrm{publ}}/m_A^2$ changes by $-2\beta_0 \ln\lambda^2\, \mathcal{M}_i^{(0)}/m_A^2$, to a relative accuracy of 7×10^{-56}. They are independent of μ_R, since their eight $\ln\mu_R^2$ terms cancel. In the original calculation, the amplitude is renormalised with Z_{GG}^2, where $Z_{GG} = Z_{a_s}$, and $C_G \propto a_s$. With the subtraction operators of Eq. (22), whose scale dependence enters through $(-\hat{s}/\mu_R^2)^{\epsilon/2}$ at one loop, renormalisation group invariance of Eq. (21) then requires

$$\frac{\partial \mathcal{M}^{(1)}}{\partial \ln \mu_R^2} = 2\beta_0\, \mathcal{M}^{(0)}. \tag{B1}$$

The first line of Eq. (24) adds this dependence, and the resulting remainders are invariant under the rescaling to 6×10^{-57}.

Scale dependence at two loops. The same ingredients determine the scale dependence at two loops. The subtraction operator $I^{(2)}$ of the original code differs from Eq. (3.4) of Ref. [7] in one respect: it does not multiply the constant

$$H_g^{(2)} = \tfrac{1}{2}\Big[C_A^2\big(-\tfrac{5}{12} - \tfrac{11}{24}\zeta_2 - \tfrac{1}{2}\zeta_3\big) + C_A n_f\big(\tfrac{29}{27} + \tfrac{\zeta_2}{12}\big) - \tfrac{1}{2}C_F n_f - \tfrac{5}{27}n_f^2\Big] \tag{B2}$$

by $(-\hat{s}/\mu_R^2)^{\epsilon}$. With this operator, one finds

$$\frac{\partial \mathcal{M}^{(2)}}{\partial \ln \mu_R^2} = 3\beta_0\, \mathcal{M}^{(1)} + \big(2\beta_1 - 8H_g^{(2)} - 12C_F n_f\big)\mathcal{M}^{(0)}, \tag{B3}$$

where $\beta_1 = \frac{34}{3}C_A^2 - 2C_F n_f - \frac{10}{3}C_A n_f$. The last term comes from the counterterm of the O_G–O_J amplitude, Eq. (25), in the renormalisation of Ref. [7]. We fitted $\partial\mathcal{M}^{(2)}/\partial\ln\mu_R^2 = a\,\beta_0\mathcal{M}^{(1)} + b\,\mathcal{M}^{(0)}$ to the series at two kinematic points, for $(N_c, n_f, m_A) = (3, 0, 300\,\mathrm{GeV})$, $(3, 5, 125\,\mathrm{GeV})$ and $(4, 2, 150\,\mathrm{GeV})$. With both lines of Eq. (24), the fit gives $a = 3$ to 2×10^{-7}, and b agrees with Eq. (B3) to 2×10^{-5}, the accuracy of the series at these velocities. The second derivative equals $6\beta_0^2\mathcal{M}^{(0)}$, the value implied by Eqs. (B1) and (B3), to 1×10^{-8}. Without the division by 64, the fit gives $a = 192$. Without the scale term of the one-loop remainders, b differs from Eq. (B3) by $6\beta_0^2\ln(m_A^2/\mathrm{GeV}^2)$. The agreement therefore also fixes the reference scale of the released one-loop remainders at $\mu_R = 1$ GeV.

Origin of the factor 64. The code of the original calculation documents this factor [7]. The two-loop renormalisation step multiplies the O_G–O_G amplitude by $64 = 8^2$, including the tree and one-loop amplitudes that enter the infrared subtraction, whereas the one-loop step does not. The pole cancellation of Ref. [7] is unaffected, since all terms of the two-loop subtraction carry the same factor.

TABLE VIII. Evaluation time for the two complete Type II coefficients at the benchmark parameters, $c = 0$ and 80-digit working precision, for the unexpanded remainders (exact) and the series. First calls start with empty caches; repeated times are medians of seven evaluations at the same point. Starting the kernel and loading the files are excluded.

	first call [s]			repeated [s]		
β_A	exact	series	ratio	exact	series	ratio
1/10	56.04	11.61	4.83	0.3422	0.0178	19.18
1/5	45.68	11.64	3.92	0.3403	0.0179	18.99
3/10	44.64	11.64	3.84	0.3485	0.0181	19.22

The code and data released with this article use the normalisation of Eq. (24).

The original output. The Mathematica output of the original calculation confirms both conversions. Its one-loop remainders, normalised to the tree amplitude, depend on μ_R as in Eq. (B1) and agree with the first line of Eq. (24) to 5×10^{-56}; the released one-loop files coincide with them at $\mu_R = 1$ GeV. For two loops, an earlier run of the renormalisation step, without the factor 64 and with the O_J contribution switched off, was evaluated at $x = 1/11$, $z = 3/11$. There the real part of the released P_1 remainder agrees to 5×10^{-31} with 64 times this result, multiplied by the tree amplitude, plus the O_G–O_J counterterm; the imaginary parts differ by $32C_F$ times the one-loop O_G–O_J amplitude, a multiple of the tree amplitude. The earlier run is not an alternative result: its renormalisation constant exceeds $Z_{GG}^2 = Z_{a_s}^2$ by a $2\beta_1 a_s^2/\epsilon$ pole and it has no O_G–O_J counterterm, so its single poles do not cancel with the IR subtraction of the released version. In the released version, the normalisation of this counterterm is fixed by the cancellation of the $C_F n_f$ single pole, and a June 2019 record of the original calculation requires the same factor in both projections at seven kinematic points. The files and a script that reproduce these comparisons are released with this article.

Appendix C: Flavour decomposition

The decomposition of Eq. (23) separates the part of each remainder that is independent of n_f from the parts proportional to n_f and n_f^2. To see whether these parts add or cancel in the complete remainder, we evaluate it at $n_f = 5$:

$$\begin{aligned}\mathcal{M}_i^{(2)}(5) &= \mathcal{M}_{i,0}^{(2)} + 5\,\mathcal{M}_{i,1}^{(2)} + 5^2\,\mathcal{M}_{i,2}^{(2)} \\ &\equiv a_0 + a_1 + a_2, \qquad i = 1, 2.\end{aligned} \tag{C1}$$

The index i is suppressed in a_r. The factors 5 and 5^2 are included in a_1 and a_2, so these are the three terms of the complete remainder rather than the flavour coefficients themselves.

Figure 6 shows $|a_r|/|\mathcal{M}_i^{(2)}(5)|$ at $c = 1/3$, $N_c = 3$ and $m_A = \mu_R = 125\,\mathrm{GeV}$. Since the terms are complex, the ratios need not add to unity; a value above unity means that a term is larger in magnitude than the full remainder and is partly cancelled by the others. The n_f

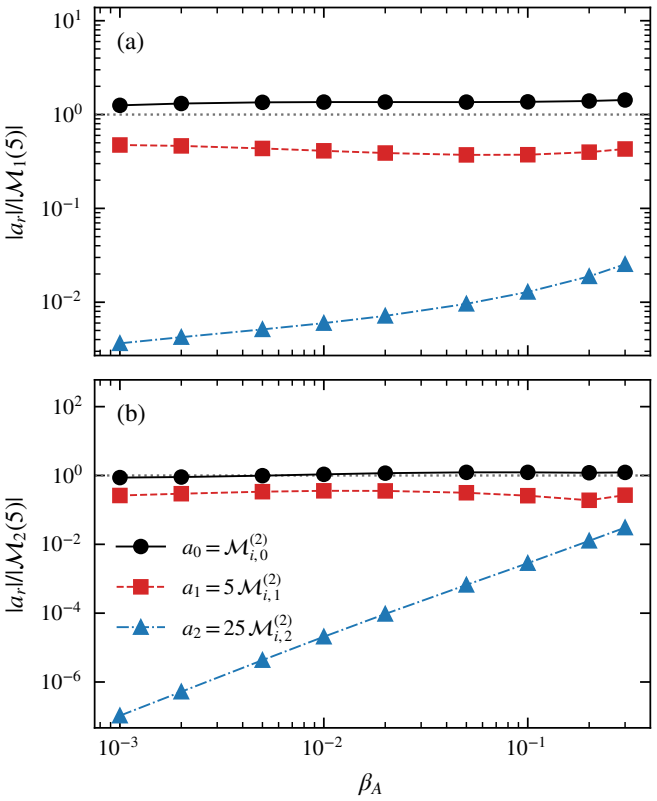


FIG. 6. Magnitudes of the three terms of Eq. (C1), divided by the magnitude of the complete coefficient: (a) $\mathcal{M}_1$, (b) $\mathcal{M}_2$, for $c = 1/3$, $m_A = \mu_R = 125\,\mathrm{GeV}$ and $N_c = 3$. The horizontal line marks unity.

independent term is the largest in both coefficients over the plotted range. The term linear in n_f is not small, and for $\mathcal{M}_1$, it partly cancels the n_f^0 term. The cancellation among the three terms can be measured by

$$1 - \frac{|a_0 + a_1 + a_2|}{|a_0| + |a_1| + |a_2|},$$

which vanishes when the terms have the same phase and approaches one when they cancel completely. For $\mathcal{M}_1$, it lies between 42.2% and 46.9% over the velocities shown, so the complete remainder is roughly half the sum of the magnitudes of its parts. This refers to the remainder, not to the cross section. The flavour terms are therefore not a small correction, and a remainder evaluated at one value of n_f cannot be rescaled to another. The relative signs can be seen in the leading power coefficients. In $\widehat{A}_{12}$, the flavour dependence enters through $\beta_0 = (11N_c - 2n_f)/3$, so there the quark and gluon contributions offset each other as in the running of the coupling; $\widehat{A}_{11}$ and $\widehat{A}_{21}$ contain $N_c a_\mathrm{A} - n_f a_\mathrm{F}$ and $4N_c - n_f$ [Eqs. (40) and (41)]. These combinations show the relative minus signs; the size of the cancellation in the complete remainder also depends on their phases and on the powers of $\mathcal{L}$ that they multiply.

Finally, the n_f^2 term of $\mathcal{M}_2$ starts at $\beta_A^2(1 - c^2)$ [Eq. (34)]. At leading power, this term is proportional to the tree amplitude, with the same factor in both tensor structures (Sec. III B), so in $\mathcal{M}_2$, it shares the centrifugal suppression of the $|J_z| = 2$ tree amplitude. This is why its relative contribution in panel (b) becomes small near threshold.

Appendix D: Evaluation time and released data

The series through β_A^6 occupy 40 kB, against 204 MB for the published two-loop files, a factor of 5061. The evaluation time was measured before the reduction of Sec. III B, with the unexpanded expressions and the series in the forms used for the comparison of Sec. III C, at $\beta_A = 1/10$, $1/5$ and $3/10$, $c = 0$, $m_A = \mu_R = 125\,\mathrm{GeV}$ and $N_c = 3$, with 80-digit working precision. On a first evaluation the series is 3.84–4.83 times faster. For repeated evaluations at the same point it is 18.99–19.22 times faster, with median 19.18; this does not correspond to a scan over new points. The two timed representations occupy 0.61 MiB and 42.77 MiB on disk, a factor of 70.4. Table VIII gives the individual measurements. The measurements used Wolfram Language 13.2.0 for Linux x86 under Ubuntu 24.04.2 LTS on an AMD Ryzen 9 7950X processor with 124.9 GiB of memory.

The released package contains the construction files, one for each flavour coefficient and power of β_A from β_A^{-4} to β_A^6; the series through β_A^6 in Wolfram Language form; the reductions of the endpoint constants; the 1080 validation records and the forward angle data; the tree-level projections; and an evaluator in Python and Wolfram Language. It also contains the Wolfram Language, C++ and Python code of the threshold construction and of the validation, the outputs of the original runs, against which the construction files and the validation records are checked, and a script that repeats the construction from the published expressions. The published expressions of Ref. [7] are downloaded and checked against checksums by a script. Each table, figure and number in this paper is generated from these data by a single command, including the check of the DPS weight in Fig. 4, the logarithms of Sec. V and the normalisation checks of App. B. A test suite checks the evaluator against the validation records. Only the construction of the series from the unexpanded expressions requires Mathematica. The series are intended for $\beta_A \leq 0.2$ at the angles tested (Sec. V C); outside this range the unexpanded remainders should be used.

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